

Notes on Abstract Algebra

Brian Zu

Following *Dummit & Foote*, 3rd edition

Contents

Chapter 0	Preliminaries	Page 2
0.1	Relations, Equivalence, and Partitions	2
0.2	Properties of Integers	3
0.3	The integers modulo n	5
Chapter 1	Introduction to Groups	Page 7
1.1	Basic Axioms and Examples	7
1.2	Dihedral Groups	15
1.3	Symmetric Group	19
	Exercises: cycles and orders — 20	
1.4	Matrix Groups	21
	Exercises: counting matrices — 22	
1.5	The Quaternion Group	23
1.6	Homomorphisms and Isomorphisms	23
	Examples of isomorphisms — 24 • Properties of homomorphisms — 27 • Groups that are not isomorphic — 28 • Constructing isomorphisms — 29	
1.7	Group Action	32
Chapter 2	Subgroups	Page 34
2.1	Subgroup definition and basic examples	34
	Verifying subgroups — 34 • The torsion subgroup — 35 • Subgroups of matrix groups and intersections — 36	
2.2	Centralizers and normalizers, stabilizers and kernels	37
	Stabilizers and kernels of group actions — 39	
2.3	Cyclic Groups and Cyclic Subgroups	43

Chapter 0

Preliminaries

0.1 Relations, Equivalence, and Partitions

Definition 0.1.1: Binary relation

Let A be a nonempty set. A **binary relation** on A is a subset $R \subseteq A \times A$. We write $a \sim b$ if $(a, b) \in R$.

Definition 0.1.2: Equivalence relation

The relation $\sim$ on A is:

1. **reflexive** if $a \sim a$ for all $a \in A$;
2. **symmetric** if $a \sim b \Rightarrow b \sim a$ for all $a, b \in A$;
3. **transitive** if $a \sim b$ and $b \sim c \Rightarrow a \sim c$ for all $a, b, c \in A$.

A relation is an **equivalence relation** if it satisfies the three properties above.

Definition 0.1.3: Equivalence class

Suppose $\sim$ defines an equivalence relation on A . The **equivalence class** of $a \in A$ is

$$\{x \in A : x \sim a\}.$$

Elements of the equivalence class of a are said to be *equivalent* to a . Any element $x \in A$ is said to be a *representative* of the class.

Definition 0.1.4: Partition

A **partition** of A is a collection $\{A_i : i \in I\}$ of nonempty subsets of A (where I is some indexing set) such that

1. $A = \bigcup_{i \in I} A_i$, and
2. $A_i \cap A_j = \emptyset$ for all $i, j \in I$ with $i \neq j$.

(Here A is the disjoint union of the sets in the partition.)

Note:-

The handwritten notes write $A_i \cap A_j \neq \emptyset$; the correct condition for a partition is $= \emptyset$.

Proposition 0.1.1 Equivalence relations and partitions

Let A be a nonempty set.

1. If $\sim$ defines an equivalence relation on A , then the set of equivalence classes of $\sim$ forms a partition of A .
2. If $\{A_i : i \in I\}$ is a partition of A , then there is an equivalence relation on A whose equivalence classes are precisely the sets $A_i, i \in I$.

0.2 Properties of Integers

Definition 0.2.1: Well ordering

$\mathbb{Z}^+$ satisfies the **well-ordering property**: every nonempty subset $A \subseteq \mathbb{Z}^+$ has a least element m , i.e. for all $x \in A$ we have $m \leq x$.

Definition 0.2.2: Divisibility

For $a, b \in \mathbb{Z}$ with $a \neq 0$, we say a **divides** b (written $a \mid b$) if there exists $d \in \mathbb{Z}$ such that $b = ad$.

Definition 0.2.3: Greatest common divisor

Let $a, b \in \mathbb{Z}$ not both zero. The **greatest common divisor** $\gcd(a, b)$ is the largest positive integer that divides both a and b . We say a and b are **relatively prime** (or coprime) if $\gcd(a, b) = 1$.

Definition 0.2.4: Least common multiple

For $a, b \in \mathbb{Z}$, the **least common multiple** $\text{lcm}(a, b)$ is the smallest positive integer that is a multiple of both a and b .

Theorem 0.2.1 Division Algorithm

Let $a, b \in \mathbb{Z}$ with $b \neq 0$. Then there exist unique $q, r \in \mathbb{Z}$ such that

$$a = qb + r, \quad 0 \leq r < |b|.$$

Theorem 0.2.2 Euclidean Algorithm

The Euclidean algorithm produces $\gcd(a, b)$ for $a, b \in \mathbb{Z} - \{0\}$. Repeatedly apply the

division algorithm:

$$\begin{aligned}
 a &= q_0b + r_0 \\
 b &= q_1r_0 + r_1 \\
 r_0 &= q_2r_1 + r_2 \\
 &\vdots \\
 r_{n-2} &= q_nr_{n-1} + r_n \\
 r_{n-1} &= q_{n+1}r_n
 \end{aligned}$$

where r_n is the last nonzero remainder. Since $(b, r_0, r_1, \dots, r_n)$ is strictly decreasing, r_n is the last such remainder. Then

$$r_n = \gcd(a, b).$$

Theorem 0.2.3 Bézout's Identity

If $a, b \in \mathbb{Z} - \{0\}$, then there exist $x, y \in \mathbb{Z}$ such that

$$\gcd(a, b) = ax + by,$$

i.e. $\gcd(a, b)$ may be written as a linear combination of a and b .

Proof: From the Euclidean algorithm we have $r_{n-2} = q_nr_{n-1} + r_n$, hence

$$r_n = r_{n-2} - q_nr_{n-1}.$$

Writing r_{n-2} and r_{n-1} in terms of their predecessors and repeatedly back-substituting, we express r_n in terms of a and b . Since $r_n = \gcd(a, b)$, the result follows. $\odot$

Definition 0.2.5: Prime number

A positive integer $p > 1$ is **prime** if its only positive divisors are 1 and p .

Theorem 0.2.4 Fundamental Theorem of Arithmetic

If $n \in \mathbb{Z}$ with $n > 1$, then there exist distinct primes $p_1, \dots, p_s$ and positive integers $a_1, \dots, a_s$ such that

$$n = p_1^{a_1} p_2^{a_2} \cdots p_s^{a_s}.$$

Moreover, if

$$n = p_1^{a_1} \cdots p_s^{a_s}, \quad b = p_1^{\beta_1} \cdots p_s^{\beta_s}$$

with all exponents ≥ 0 , then

$$\gcd(a, b) = p_1^{\min(a_1, \beta_1)} \cdots p_s^{\min(a_s, \beta_s)}, \quad \text{lcm}(a, b) = p_1^{\max(a_1, \beta_1)} \cdots p_s^{\max(a_s, \beta_s)}.$$

Definition 0.2.6: Euler φ -function

For $n \in \mathbb{Z}^+$, let $\varphi(n)$ be the number of positive integers $\leq n$ that are relatively prime to n :

$$\varphi(n) = \#\{k : 1 \leq k \leq n, \gcd(k, n) = 1\}.$$

Example 0.2.1 ($\varphi(12) = 4$)

$\varphi(12) = 4$, since 1, 5, 7, 11 are relatively prime to 12 and ≤ 12 .

Theorem 0.2.5 Euler φ -function

- For a prime p , $\varphi(p) = p - 1$, and for all $a \geq 1$,

$$\varphi(p^a) = p^a - p^{a-1} = p^{a-1}(p - 1).$$

- φ is **multiplicative** in the sense that

$$\varphi(ab) = \varphi(a)\varphi(b) \quad \text{if } \gcd(a, b) = 1.$$

- If $n = p_1^{a_1} \cdots p_s^{a_s}$, then

$$\varphi(n) = \varphi(p_1^{a_1}) \cdots \varphi(p_s^{a_s}) = p_1^{a_1-1}(p_1 - 1) \cdots p_s^{a_s-1}(p_s - 1).$$

0.3 The integers modulo n

Definition 0.3.1: Congruence modulo n

Define a relation on $\mathbb{Z}$ by

$$a \sim b \quad \text{iff} \quad n \mid (b - a).$$

For any $k \in \mathbb{Z}$ the equivalence class of k is denoted $\bar{k}$, and is called the **congruence class** (or **residue class**) of k modulo n :

$$\bar{k} = \{k + tn : t \in \mathbb{Z}\}.$$

There are n distinct equivalence classes modulo n , namely

$$\bar{0}, \bar{1}, \dots, \overline{n-1}.$$

The set of equivalence classes is denoted $\mathbb{Z}/n\mathbb{Z}$.

Definition 0.3.2: Operations on $\mathbb{Z}/n\mathbb{Z}$

For $\bar{a}, \bar{b} \in \mathbb{Z}/n\mathbb{Z}$,

$$\bar{a} + \bar{b} = \overline{a + b}, \quad \bar{a} \cdot \bar{b} = \overline{a \cdot b}.$$

Theorem 0.3.1 Well-defined operations

Addition and multiplication on $\mathbb{Z}/n\mathbb{Z}$ are **well defined**, i.e. they do not depend on the choice of representatives for the classes involved. Explicitly, if $a_1, a_2, b_1, b_2 \in \mathbb{Z}$ with $\bar{a}_1 = \bar{b}_1$ and $\bar{a}_2 = \bar{b}_2$, then

$$\overline{a_1 + a_2} = \overline{b_1 + b_2}, \quad \overline{a_1 \cdot a_2} = \overline{b_1 \cdot b_2}.$$

Proof: Suppose $a_1 \equiv b_1 \pmod{n}$ and $a_2 \equiv b_2 \pmod{n}$. Then there exist $s, t \in \mathbb{Z}$ such that

$$a_1 = b_1 + sn, \quad a_2 = b_2 + tn.$$

For addition,

$$a_1 + a_2 = b_1 + b_2 + (s + t)n \implies a_1 + a_2 \equiv b_1 + b_2 \pmod{n}.$$

For multiplication,

$$a_1 a_2 = (b_1 + sn)(b_2 + tn) = b_1 b_2 + n(b_1 t + b_2 s + stn),$$

so $a_1 a_2 \equiv b_1 b_2 \pmod{n}$. ☺

Definition 0.3.3: Units of $\mathbb{Z}/n\mathbb{Z}$

An important subset of $\mathbb{Z}/n\mathbb{Z}$ is

$$(\mathbb{Z}/n\mathbb{Z})^\times = \{\bar{a} \in \mathbb{Z}/n\mathbb{Z} : \exists \bar{b} \in \mathbb{Z}/n\mathbb{Z} \text{ with } \bar{a} \cdot \bar{b} = \bar{1}\}.$$

Proposition 0.3.1 Unit criterion

$$(\mathbb{Z}/n\mathbb{Z})^\times = \{\bar{a} \in \mathbb{Z}/n\mathbb{Z} : \gcd(a, n) = 1\}.$$

Example 0.3.1 (Units modulo 9)

For $n = 9$,

$$(\mathbb{Z}/9\mathbb{Z})^\times = \{\bar{1}, \bar{2}, \bar{4}, \bar{5}, \bar{7}, \bar{8}\}.$$

Since $\gcd(3, 9) = 3 \neq 1$, we have $\bar{3} \notin (\mathbb{Z}/9\mathbb{Z})^\times$.

Chapter 1

Introduction to Groups

1.1 Basic Axioms and Examples

Definition 1.1.1: Binary operation

A **binary operation** $*$ on a set G is a function $*$: $G \times G \rightarrow G$. For all $a, b \in G$ we write $a * b$ for $*(a, b)$.

Definition 1.1.2: Associativity

$*$ on G is **associative** if for all $a, b, c \in G$,

$$a * (b * c) = (a * b) * c.$$

Definition 1.1.3: Commutativity

$*$ on G is **commutative** if for all $a, b \in G$,

$$a * b = b * a.$$

Example 1.1.1 (Subtraction is not commutative)

(Usual subtraction) $-$ is a **noncommutative** binary operation on $\mathbb{Z}$. For instance

$$(-1) - 1 = -2, \quad 1 - (-1) = 2,$$

so $(-1) - 1 \neq 1 - (-1)$.

Note:-

The handwritten notes say “commutative” here, but the computation shows subtraction is not commutative.

Definition 1.1.4: Closure under a subset

Let $H \subseteq G$. If for all $x, y \in H$ we have $x * y \in H$, then H is said to be **closed under $*$** .

Definition 1.1.5: Group

A **group** is an ordered pair $(G, *)$ where G is a set and $*$ is a binary operation on G satisfying the following axioms:

1. **Associativity:** $a * (b * c) = (a * b) * c$ for all $a, b, c \in G$.
2. **Identity:** there exists an element $e \in G$, called the **identity** of G , such that $e * a = a * e = a$ for all $a \in G$.
3. **Inverses:** for each $a \in G$ there exists an element $a^{-1} \in G$, called the **inverse** of a , such that $a * a^{-1} = a^{-1} * a = e$.

Definition 1.1.6: Abelian group

The group $(G, *)$ is **abelian** (or **commutative**) if $a * b = b * a$ for all $a, b \in G$.

Definition 1.1.7: Finite group

$(G, *)$ is a **finite group** if G is a finite set.

Example 1.1.2 (Groups under $+$ and $\times$)

1. $\mathbb{Z}, \mathbb{Q}, \mathbb{R}, \mathbb{C}$ are groups under $+$, with identity $e = 0$ and $a^{-1} = -a$ for all a .
2. $\mathbb{Q} - \{0\}, \mathbb{R} - \{0\}, \mathbb{C} - \{0\}, \mathbb{Q}^+, \mathbb{R}^+$ are groups under $\times$, with identity $e = 1$ and $a^{-1} = \frac{1}{a}$ for all a . However, $\mathbb{Z} - \{0\}$ is **not** a group under $\times$: taking $a = 2$, there is no $a^{-1} \in \mathbb{Z}$ with $a \cdot a^{-1} = 1$, since $a^{-1} = \frac{1}{2} \notin \mathbb{Z}$.
3. For a vector space V , $(V, +)$ is an abelian group.
4. For $n \in \mathbb{Z}^+$, $\mathbb{Z}/n\mathbb{Z}$ is an abelian group under the operation $+$ of residue classes. The identity is $\bar{0}$, and the inverse of $\bar{a}$ is $\overline{-a}$.
5. For $n \in \mathbb{Z}^+$, $(\mathbb{Z}/n\mathbb{Z})^\times$ is an abelian group under multiplication of residue classes. The identity is $\bar{1}$; indeed, by definition, for every $\bar{a} \in (\mathbb{Z}/n\mathbb{Z})^\times$ there is a $\bar{c} \in \mathbb{Z}/n\mathbb{Z}$ with $\bar{a} \cdot \bar{c} = \bar{1}$, i.e. every element has an inverse.
6. If $(A, *)$ and $(B, \diamond)$ are groups, then the **direct product** $A \times B$ is a group under the componentwise operation

$$(a_1, b_1)(a_2, b_2) = (a_1 * a_2, b_1 \diamond b_2),$$

where $A \times B = \{(a, b) : a \in A, b \in B\}$.

Proposition 1.1.1 Basic properties

Let G be a group under $*$.

1. The identity of G is unique.

2. For each $a \in G$, the inverse of a is unique.
3. For all $a \in G$, $(a^{-1})^{-1} = a$.
4. For all $a, b \in G$, $(a * b)^{-1} = b^{-1} * a^{-1}$.
5. (**Generalized associativity**) The product of n factors is independent of the way it is parenthesized; this is proved by induction on the number of factors n .

Proof: (1) If f and g are both identities, then by axiom (ii),

$$f * g = f = g * f = g,$$

so $f = g$.

(2) Suppose b and c are both inverses of a , and let e be the identity. Then

$$\begin{aligned} c &= c * e &= c * (a * b) &\text{(since } e = a * b\text{)} \\ &= (c * a) * b &= e * b &\text{(since } c * a = e\text{)} \\ &= b. \end{aligned}$$

Hence $b = c$.

(3) Since $a * a^{-1} = e$ and $a^{-1} * a = e$, the element a is an inverse of a^{-1} ; by uniqueness (2), $(a^{-1})^{-1} = a$.

(4) Let $c = (a * b)^{-1}$, so that $(a * b) * c = e$. Then, using associativity,

$$\begin{aligned} a * (b * c) &= e \\ a^{-1} * (a * (b * c)) &= a^{-1} * e \\ (a^{-1} * a) * (b * c) &= a^{-1} \\ e * (b * c) &= a^{-1} \\ b * c &= a^{-1}. \end{aligned}$$

Now multiply on the left by b^{-1} :

$$\begin{aligned} b^{-1} * (b * c) &= b^{-1} * a^{-1} \\ (b^{-1} * b) * c &= b^{-1} * a^{-1} \\ c &= b^{-1} * a^{-1}. \end{aligned}$$

(5) By induction on the number of factors n . The cases $n = 2$ and $n = 3$ are immediate: $a_1 * a_2 = a_1 * a_2$ and $(a_1 * a_2) * a_3 = a_1 * (a_2 * a_3)$ by associativity. For the induction step, assume the statement is true for all $< n$ factors. Let

$$L_n := ((a_1 * a_2) * a_3) * \cdots * a_n$$

be the standard left-associated product; we show every bracketing equals L_n . A fully parenthesized product of $a_1, \dots, a_n$ splits at its final operation into two blocks

$$(a_1 * \cdots * a_k) * (a_{k+1} * \cdots * a_n) = X * Y$$

for some $1 \leq k < n$. By induction $X = L_k$, and Y may be replaced by its left-associated form (it has fewer than n factors). Using ordinary associativity repeatedly to move the parentheses to the left,

$$L_k * (a_{k+1} * \cdots * a_n) = ((a_1 * a_2) * \cdots * a_k) * a_{k+1} * \cdots * a_n = L_n.$$

Hence every bracketing equals L_n . ☺

Note:-**Notation**

For $a, b \in G$ we write $a * b$ simply as ab (or $a \cdot b$). For an abstract group G , the identity is denoted by 1. For $x \in G$ and $n \in \mathbb{Z}^+$, since the product $x \cdot x \cdots x$ (over n terms) is independent of bracketing, we denote it x^n ; similarly $x^{-1} \cdots x^{-1} = x^{-n}$, and $x^0 = 1$ is the identity of G .

This notation is only for abstract groups. For a specific group we must use its given operation; for example, when working with $+$, the inverse a^{-1} is written $-a$ and the identity is 0.

Proposition 1.1.2 Cancellation laws

Let G be a group and $a, b, c \in G$. The equations $ax = b$ and $ya = b$ have unique solutions $x, y \in G$. In particular, the left and right **cancellation laws** hold in G :

1. if $au = av$ then $u = v$;
2. if $ub = vb$ then $u = v$.

Proof: We solve $ax = b$ by multiplying both sides on the left by a^{-1} :

$$a^{-1}ax = a^{-1}b \implies x = a^{-1}b.$$

Then x is unique since a^{-1} is unique. Similarly, if $ya = b$ then $y = ba^{-1}$. If $au = av$, multiplying on the left by a^{-1} gives $a^{-1}au = a^{-1}av$, i.e. $u = v$; the right cancellation law is analogous. ☺

Example 1.1.3 (Solving in $(\mathbb{Z}, +)$)

Let $a = 5$, $b = 12$. To solve $5 + x = 12$, add the inverse of 5, namely -5 , to both sides:

$$-5 + 5 + x = -5 + 12 \implies x = -5 + 12 = 7,$$

and such an x is unique. Indeed, if $5 + u = 5 + v$, then

$$-5 + 5 + u = -5 + 5 + v \implies 0 + u = 0 + v \implies u = v.$$

The right-cancellation case is similar.

Corollary 1.1.1 Inverse criterion

If a is any element of G and for some $b \in G$ we have $ab = e$ or $ba = e$, then $b = a^{-1}$.

Proof: Viewing $ab = e$ as the equation $ax = e$ with $x = b$, and noting that the equation has a unique solution (by the previous Prop) which is a^{-1} , we conclude $b = a^{-1}$. The case $ba = e$ is analogous. ☺

Definition 1.1.8: Order of an element

Let G be a group and $x \in G$. The **order** of x is the smallest positive integer n such that $x^n = 1$; we write $|x| = n$ and say x has order n . If no positive power of x equals the identity, then x is said to have **infinite order** (and we write $|x| = \infty$).

Example 1.1.4 (Orders of elements)

1. An element of a group has order 1 if and only if it is the identity, i.e. $|1| = 1$.
2. In the additive groups $\mathbb{Z}, \mathbb{Q}, \mathbb{R}, \mathbb{C}$, every nonzero element has infinite order (these are ordered, and repeated addition only makes the element larger).
3. In the multiplicative groups $\mathbb{R} - \{0\}$ or $\mathbb{Q} - \{0\}$, the element -1 has order 2, and every other non-identity element has infinite order.
4. In the additive group $\mathbb{Z}/5\mathbb{Z}$, the element $\bar{1}$ has order 5: $\bar{1} \neq \bar{0}$, but

$$\bar{1} + \bar{1} + \bar{1} + \bar{1} + \bar{1} = \bar{5} = \bar{0},$$

which is the identity.

5. In the multiplicative group $(\mathbb{Z}/7\mathbb{Z})^\times$, the powers of $\bar{2}$ are $\bar{2}$, $\bar{2}^2 = \bar{4}$, and $\bar{2}^3 = \bar{8} = \bar{1}$, the identity; hence $\bar{2}$ has order 3.

Definition 1.1.9: Multiplication table

Let $G = \{g_1, g_2, \dots, g_n\}$ be a finite group. The **multiplication table** (or **group table**) of G is the $n \times n$ matrix whose (i, j) -entry is the group element $g_i \cdot g_j$.

Question 1: Exercise 1

- (a) Subtraction is **not** associative. For instance,

$$a - (b - c) = a - b + c, \quad (a - b) - c = a - b - c,$$

which differ in general.

- (b) Define $a * b = a + b + ab$. Then $*$ is **associative**:

$$\begin{aligned} a * (b * c) &= a * (b + c + bc) = a + b + c + bc + a(b + c + bc) \\ &= a + b + c + bc + ab + ac + abc, \\ (a * b) * c &= (a + b + ab) * c = a + b + ab + c + (a + b + ab)c \\ &= a + b + ab + c + ac + bc + abc = a * (b * c). \end{aligned}$$

- (c) Define $a * b = \frac{a+b}{2}$. Then $*$ is **not** associative:

$$\begin{aligned} a * (b * c) &= a * \frac{b+c}{2} = \frac{a + \frac{b+c}{2}}{2} = \frac{2a + b + c}{4}, \\ (a * b) * c &= \frac{a+b}{2} * c = \frac{\frac{a+b}{2} + c}{2} = \frac{a + b + 2c}{4}, \end{aligned}$$

which differ in general.

(d) Define $(a, b) * (c, d) = (ad + bc, bd)$. Then $*$ is **associative**:

$$\begin{aligned} (a, b) * ((c, d) * (e, f)) &= (a, b) * (cf + de, df) = (a \cdot df + b(cf + de), b \cdot df) \\ &= (adf + bcf + bde, bdf), \\ ((a, b) * (c, d)) * (e, f) &= (ad + bc, bd) * (e, f) = ((ad + bc)f + bd \cdot e, bd \cdot f) \\ &= (adf + bcf + bde, bdf). \end{aligned}$$

They agree, so $*$ is associative.

(e) Define $a * b = \frac{a}{b}$. Then $*$ is **not** associative:

$$\begin{aligned} a * (b * c) &= a * (b/c) = \frac{a}{b/c} = \frac{ac}{b}, \\ (a * b) * c &= (a/b) * c = \frac{a/b}{c} = \frac{a}{bc}, \end{aligned}$$

which differ in general.

Question 2: Associativity in $\mathbb{Z}/n\mathbb{Z}$

1. Let $[a], [b], [c] \in \mathbb{Z}/n\mathbb{Z}$. Since $[a] + [b] = [a + b]$, we have

$$\begin{aligned} ([a] + [b]) + [c] &= [a + b] + [c] = [(a + b) + c], \\ [a] + ([b] + [c]) &= [a] + [b + c] = [a + (b + c)]. \end{aligned}$$

Because addition in $\mathbb{Z}$ is associative, $[(a + b) + c] = [a + (b + c)]$; hence $+$ is associative on $\mathbb{Z}/n\mathbb{Z}$.

2. Since $[a] \cdot [b] = [ab]$, and multiplication in $\mathbb{Z}$ is associative, $\cdot$ is also associative on $\mathbb{Z}/n\mathbb{Z}$.
3. $\mathbb{Z}/n\mathbb{Z}$ is **not** a group under multiplication: the class $[0]$ has no multiplicative inverse, since for every $[a] \in \mathbb{Z}/n\mathbb{Z}$ we have $[0] \cdot [a] = [0] \neq [1]$.

Question 3: Exercise 6 — subsets of $\mathbb{Q}$

- (a) For the set of rational numbers: $\frac{1}{3} + \frac{1}{3} = \frac{2}{3}$ (closure); the inverse $\frac{-a}{b}$ of $\frac{a}{b}$ is in the set; $0 = \frac{0}{1}$ is the identity; and associativity holds, $\frac{a}{b} + \left(\frac{c}{d} + \frac{e}{f}\right) = \left(\frac{a}{b} + \frac{c}{d}\right) + \frac{e}{f}$. Hence it is a group under $+$.
- (b) $\frac{1}{2} + \frac{1}{2} = 1$ is **not closed**: $\frac{2}{2}$ reduces to $\frac{1}{1}$, which is not in the set, and there is no identity. Not a group.
- (c) $\frac{3}{4} + \frac{3}{4} = \frac{3}{2} > 1$, so the set is not closed. Not a group.
- (d) $\frac{a}{2} + \frac{b}{2} = \frac{a+b}{2}$ (closed); $\frac{0}{1}$ is the identity; associativity holds for rational addition. Hence it is a group under $+$.
- (e) $\frac{a}{b}$ has inverse $\frac{-a}{b}$ (and similarly $\frac{-c}{d}$), which lies in the set; hence a group under $+$.
- (f) $\frac{1}{2} + \frac{1}{2}$ is not in the set, so it is not closed. Not a group.

Question 4: Exercise 7 — $[0, 1)$ modulo 1

Let $G = \{x \in \mathbb{R} : 0 \leq x < 1\}$, and define

$$x * y = \{x + y\},$$

where $\{x\} = x - \lfloor x \rfloor$ is the fractional part of x . Since $0 \leq x + y - \lfloor x + y \rfloor < 1$, the operation is **well defined**.

Claim 1.1.1 Association

$$(x * y) * z = \{(x + y) + z\} = \{x + y + z\},$$

$$x * (y * z) = \{x + (y + z)\} = \{x + y + z\}$$

The identity is 0, since $0 * x = \{0 + x\} = x$ for every $x \in G$. If $x = 0$ its inverse is 0; if $0 < x < 1$, then

$$x * (1 - x) = \{x + 1 - x\} = \{1\} = 0,$$

so $x^{-1} = 1 - x$.

Claim 1.1.2 Commutativity

$x * y = \{x + y\} = \{y + x\} = y * x$, so G is abelian under $*$.

Question 5: Exercise 8 — roots of unity

(a) Let $G = \{z \in \mathbb{C} : z^n = 1 \text{ for some } n \in \mathbb{Z}^+\}$. We show G is a group under multiplication.

① **Closure:** let $z, w \in G$, so $z^m = 1$ and $w^n = 1$ for some m, n . Then

$$(zw)^{mn} = z^{mn}w^{mn} = (z^m)^n(w^n)^m = 1,$$

so $zw \in G$.

② **Identity:** $1 \in G$, since $1^1 = 1$.

③ **Inverse:** if $z^n = 1$, then $z^{n-1} = z^{-1}$. Indeed $z^n = z \cdot z^{n-1} = 1$, so z^{n-1} is the inverse of z .

④ **Associativity:** inherited from complex multiplication.

(b) Let $z, w \in G$; we need $zw \in G$. Take $z = 1$, $w = -1$. Both lie in G since $1^1 = 1$ and $(-1)^2 = 1$. But $1 + (-1) = 0 \notin G$, so (under the operation $+$) the set fails closure.

Question 6: Exercise 9 — $\mathbb{Q}(\sqrt{2})$

(a) Let $G = \{a + b\sqrt{2} : a, b \in \mathbb{Q}\}$. Under addition:

① **Closure:** $(a + b\sqrt{2}) + (c + d\sqrt{2}) = (a + c) + (b + d)\sqrt{2}$, with $a + c \in \mathbb{Q}$ and $b + d \in \mathbb{Q}$.

② **Identity:** $(a + b\sqrt{2}) + 0 = a + b\sqrt{2}$, so 0 is the identity.

③ **Inverse:** $(a + b\sqrt{2}) + (-a - b\sqrt{2}) = 0$, so $-a - b\sqrt{2}$ is the inverse.

④ **Associativity:** inherited from addition in $\mathbb{R}$.

(b) For $G = (\mathbb{Q}(\sqrt{2}))^\times$, under multiplication:

① **Closure:**

$$(a + b\sqrt{2})(c + d\sqrt{2}) = ac + ad\sqrt{2} + bc\sqrt{2} + 2bd = (ac + 2bd) + (ad + bc)\sqrt{2}.$$

② **Identity:** $1 + 0\sqrt{2} = 1$.

③ **Inverse:** let $x = a + b\sqrt{2} \neq 0$. Then

$$\frac{1}{x} = \frac{1}{a + b\sqrt{2}} = \frac{a - b\sqrt{2}}{a^2 - 2b^2}.$$

We need $a^2 - 2b^2 \neq 0$. Indeed, if $a^2 = 2b^2$ and $b \neq 0$, then $\left(\frac{a}{b}\right)^2 = 2$, i.e. $\frac{a}{b} = \sqrt{2}$, contradicting the irrationality of $\sqrt{2}$.

Question 7: Order of elements in $\mathbb{Z}/12\mathbb{Z}$

In the additive group $\mathbb{Z}/12\mathbb{Z}$ the order of $\bar{k}$ is the smallest positive integer n such that $n\bar{k} = \bar{0}$, i.e. the smallest n with $12 \mid nk$.

Element	Order
$\bar{1}$	12
$\bar{2}$	6
$\bar{3}$	4
$\bar{4}$	3
$\bar{5}$	12
$\bar{6}$	2
$\bar{7}$	12
$\bar{8}$	3
$\bar{9}$	4
$\bar{10}$	6
$\bar{11}$	12

Note:-

These are the orders $|\bar{k}| = 12/\gcd(k, 12)$ obtained from the criterion above; they differ from the handwritten table, which I read as inconsistent with the stated criterion.

Claim 1.1.3 General formula

Let $d = \gcd(12, k)$ and write $12 = d \cdot 12'$, $k = d \cdot k'$ with $\gcd(12', k') = 1$. Then

$$12 \mid nk \iff d \cdot 12' \mid n \cdot d \cdot k' \iff 12' \mid nk'.$$

Since $\gcd(12', k') = 1$, the condition $12' \mid nk'$ is equivalent to $12' \mid n$. Hence the smallest positive such n is $n = 12'$, i.e.

$$|\bar{k}| = n = \frac{12}{d} = \frac{12}{\gcd(12, k)}.$$

1.2 Dihedral Groups

Definition 1.2.1

Think of a symmetry as a movement of a polygon when it lands on top of itself. For a regular n -gon the symmetries are the **rotations and reflections** that preserve the polygon. The set of all these symmetries is called D_{2n} .

Definition 1.2.2: Size of D_{2n}

For a regular n -gon there are n rotations and n reflections, so the total number of symmetries is $2n$, i.e. $|D_{2n}| = 2n$.

Definition 1.2.3: Description by permutations

Each symmetry s can be described uniquely by a permutation σ of $\{1, 2, 3, \dots, n\}$, where if the symmetry s puts vertex i in the place of the vertex j originally, then σ is the permutation sending i to j (i.e. $\sigma(i) = j$).

Example 1.2.1 (Rotation by $\frac{2\pi}{n}$)

For example, if s is a rotation of $\frac{2\pi}{n}$ clockwise about the center of the n -gon, then σ is the permutation sending $i \mapsto i + 1$ for $1 \leq i \leq n - 1$, and $\sigma(n) = 1$.

Definition 1.2.4: Group structure

We make D_{2n} into a group by defining st (for $s, t \in D_{2n}$) to be the symmetry obtained by first applying t and then s to the n -gon. If s, t effect the permutations σ, τ respectively, then st effects the permutation $\sigma \circ \tau$.

The binary operation of D_{2n} is **associative**, since composition of functions is associative. The **identity** is the identity symmetry, denoted 1. The **inverse** of $s \in D_{2n}$ is the symmetry which reverses all the rigid motions of s .

Question 8: Orders in D_6

Recall that the order of an element $g \in G$ is the smallest positive integer m with $g^m = e$; we write $|g| = m$. For

$$D_6 = D_{2 \cdot 3} = \{1, r, r^2, s, sr, sr^2\},$$

Element	Order
1	1
r	3
r^2	3
s	2
sr	2
sr^2	2

Claim 1.2.1

If x is not a power of r , then $x = sr^k$ for some k , and

$$rx = r(sr^k) = sr^{-1}r^k = sr^{k-1} = xr^{-1},$$

so $rx = xr^{-1}$.

Claim 1.2.2

If x is not a power of r , then $x = sr^k$ and using $sr^k s = r^{-k}$,

$$x^2 = sr^k sr^k = (sr^k s)r^k = r^{-k}r^k = 1.$$

Hence every reflection has order 2.

Question 9: The center of D_{2n}

Claim 1.2.3

$D_{2n} = \langle s, sr \rangle$: since $s(sr) = r$, we have $r \in \langle s, sr \rangle$, and $s \in \langle s, sr \rangle$, so $\langle s, sr \rangle$ contains s and r , hence all of D_{2n} .

Suppose $x \in D_{2n}$ commutes with every element, and (for n even) write $n = 2k$ so that

$z = r^k$ is the half-turn, satisfying $z^2 = 1$, $|z| = 2$.

Case 1: $x = r^i$. Then x commutes with r , and

$$r^i s = s r^i = r^{-i} s \implies r^i = r^{-i} \implies r^{2i} = 1.$$

Since r has order n , $r^{2i} = 1$ forces $n \mid 2i$. With $0 \leq i < n$ this gives $i = 0$, together with $i = n/2$ when n is even. Hence $x = 1$ always, and $x = r^{n/2} = z$ when n is even.

Case 2: $x = s r^k$ (not a power of r). Then $rx = x r^{-1}$ (shown earlier). If x commutes, $rx = xr$, hence

$$x r^{-1} = x r \implies r^{-1} = r \implies r^2 = 1,$$

so r has order 2. But r has order n ; for $n \neq 2$ this is impossible.

Therefore

$$Z(D_{2n}) = \begin{cases} \{1\}, & n \text{ odd}, \\ \{1, r^{n/2}\}, & n \text{ even}. \end{cases}$$

Question 10: More on the center

(5) Let $x \neq 1$ be a non-identity element of D_{2n} that commutes with every element (so n is odd). Then $x = r^i$ or $x = s r^i$.

- If $x = r^i$: $r^i s = s r^i = r^{-i} s$, so $r^i = r^{-i}$ and $r^{2i} = 1$. Since r has order n with n odd, $n \mid i$, so $r^i = 1$ — contradicting $x \neq 1$.
- If $x = s r^i$: by the earlier result $rx = x r^{-1}$; if x commutes then $rx = xr = x r^{-1}$, so $r^2 = 1$, contradicting $|r| = n \geq 3$.

Hence for odd n , the identity is the only element commuting with everything.

(6) In a group with generators x, y : $x^{-1} = (xy)^{-1} = y^{-1}x^{-1} = yx$, and $tx = xt^{-1}$, which matches the dihedral relation $rs = sr^{-1}$ (with $x \leftrightarrow r$ and $y \leftrightarrow s$).

Question 11: Generators and the subgroup $\langle r \rangle$

(7) Put $a = s$ and $b = sr$. Then

$$a^2 = s^2 = 1, \quad b^2 = s r s r = s(rs)r = s(sr^{-1})r = s^2 r^{-1} r = 1.$$

Since $ab = s(sr) = r$,

$$(ab)^n = r^n = 1.$$

Moreover we can recover r, s from a, b : $s = a$ and $r = ssr = ab$, so a, b generate the same group as r, s . Finally

$$rs = (ab)a = aba, \quad sr^{-1} = a(ab)^{-1} = ab^{-1}a^{-1} = aba$$

(because $b^{-1} = b$ and $a^{-1} = a$, as $|a| = |b| = 2$). Hence $rs = sr^{-1}$.

(8) $\langle r \rangle = \{r^m : m \in \mathbb{Z}\}$. Since $r^n = 1$,

$$\langle r \rangle = \{1, r, \dots, r^{n-1}\}, \quad |\langle r \rangle| = n.$$

Question 12: Orders of symmetry groups and presentations

- (9) A tetrahedron has 4 triangular faces, each having 3 vertices; so $|G| = 4 \cdot 3 = 12$.
 (10) A cube has 6 faces, each having 4 vertices; so $|G| = 4 \cdot 6 = 24$.
 (14) $\mathbb{Z} = \langle 1 \rangle$.
 (15) $\langle 1 \mid 1^n = 1 \rangle$ (a presentation of a cyclic group).
 (16) **General method** for proving something is the dihedral group D_{2n} : given a set $\langle a, b \mid \sim \rangle$ (where $\sim$ is the set of relations), and recalling

$$D_{2n} = \langle r, s \mid r^n = s^2 = 1, rs = sr^{-1} \rangle,$$

we need only substitute $r = a$ (where a is “rotation-like”) and $s = b$ (where b is “reflection-like”).

Example 1.2.2 (D_4 from a presentation)

Show that $\langle x_1, y_1 \mid x_1^2 = y_1^2 = (x_1 y_1)^2 = 1 \rangle$ is the dihedral group D_4 . Put $x_1 = r$ and $y_1 = s$. Then $r^n = r^2 = x_1^2 = 1$ and $s^2 = y_1^2 = 1$ (by definition). Since $(x_1 y_1)^2 = 1$,

$$x_1 y_1 = (x_1 y_1)^{-1} = y_1^{-1} x_1^{-1}.$$

As $x_1^2 = y_1^2 = 1$, we have $x_1^{-1} = x_1$ and $y_1^{-1} = y_1$, so

$$x_1 y_1 = y_1 x_1 = y_1 x_1^{-1} \implies rs = sr^{-1}.$$

Hence the relations match, and the presentation is D_4 .

Question 13: 17(a) — the group X_{2n}

Since $xy = yx^2$, multiplying on the left by y gives

$$y(xy) = y(yx^2) = y^2 x^2 = x^2,$$

i.e. $yxy = x^2$. Multiplying on both sides by y ,

$$x = y(yxy)y = yx^2 y.$$

Now $yx^2 y = (yx)(xy) = x^2 x^2 = x^4$ (using $yx = x^2 y$ and $xy = yx^2$), so $x = x^4$ and hence $x^3 = 1$. Therefore $x^{-1} = x^2$, and

$$xy = yx^2 = yx^{-1}.$$

Thus X_{2n} has the same presentation as $D_6 = \langle r, s \mid r^3 = s^2 = 1, rs = sr^{-1} \rangle$, by letting $x = r$ and $y = s$.

Question 14: 17(b) and 18

(17b) If $\gcd(3, n) = 1$, then since $x^3 = 1$ and $x^n = 1$, the order of x divides $\gcd(3, n) = 1$, so $x = 1$. Then $X_{2n} = \langle x, y \mid x = y^2 = 1, xy = yx^2 \rangle$ has only two elements, namely $\{1, y\}$, so $|X_{2n}| = 2$.

(18a) Reminder: $X = \langle u, v \mid u^4 = v^3 = 1, uv = v^2u \rangle$; here $v^3 = 1$, so $v^{-1} = v^2$.

(18b) We show u^3 is central. Clearly u^3 commutes with u ; it remains to show u^3 commutes with v . From $uv = v^2u$ we get

$$(v^2u^2)(uv) = (uv)^2 = uvuv = u(vu)v,$$

and since $vu = v^2u$ the relation $vuv = u^2$ holds, whence $u^3v = vu^3$. Thus u^3 commutes with v , so $u^3 \in Z(X)$.

(18c) Since $u^4 = 1$ and $(u^3)^3 = u^9 = u$, and by (b) v commutes with u^3 , it follows that v commutes with u .

(18d) $uv = v^2u^2$; by (c), $uv = vu = v^2u^2$.

(18e) From $u^4v^3 = 1$,

$$1 = u^4v^3 = u^3(uv)v^2 = u^3v^2 = u^2(uv)v = u^2v = u(uv) = u,$$

so $u = 1$. Then $uv = 1$ forces $v = 1$; by definition X is generated by u, v , so $X = 1$ is trivial.

1.3 Symmetric Group

Example 1.3.1

Let Ω be any nonempty set and let S_Ω be the set of all bijections from Ω to itself. Then S_Ω is a group under function composition $\circ$:

- **Associativity:** since function composition is associative, $\circ$ is associative.
- **Identity:** the permutation 1 defined by $1(\alpha) = \alpha$ for all $\alpha \in \Omega$.
- **Inverse:** for every permutation σ there is a two-sided inverse $\sigma^{-1} : \Omega \rightarrow \Omega$.

Thus all the group axioms are satisfied by $(S_\Omega, \circ)$.

Definition 1.3.1: Symmetric group

The group $(S_\Omega, \circ)$ is called the **symmetric group** on the set Ω .

Definition 1.3.2: Symmetric group of degree n

If $\Omega = \{1, 2, 3, \dots, n\}$, then the symmetric group is denoted S_n and is called the **symmetric group of degree n** .

Theorem 1.3.1 Order of S_n

$$|S_n| = n!.$$

Proof: $\sigma(1)$ may be any of the n elements, $\sigma(2)$ any element except $\sigma(1)$, and so on. Hence the number of possible injective functions from $\{1, \dots, n\}$ to itself is

$$n(n-1)(n-2) \cdots 2 \cdot 1 = n!.$$

So there are $n!$ elements in S_n . ☺

Definition 1.3.3: Cycle

A **cycle** is a string of integers which represents the element of S_n which cyclically permutes these integers (and fixes all the other integers).

Example 1.3.2 (Cycle notation)

$(a_1 a_2 \cdots a_m)$ is the permutation which sends a_i to a_{i+1} for $1 \leq i \leq m-1$, and a_m to a_1 .

Definition 1.3.4: Cycle decomposition

Generally, for each $\sigma \in S_n$ the numbers 1 to n are rearranged and grouped into cycles of the form

$$(a_1 a_2 \cdots a_m)(a_{m+1} a_{m+2} \cdots a_{m_2}) \cdots (a_{m_{k-1}} \cdots a_{m_k}),$$

i.e. σ is a product of disjoint cycles. The product of all the cycles is called the **cycle decomposition** of σ .

1.3.1 Exercises: cycles and orders**Example 1.3.3** (1.)

Let $\sigma = (135)(24)$ and $\tau = (15)(23)$. Then

$$\sigma^2 = (153), \quad \sigma\tau = (2534), \quad \tau\sigma = (1243), \quad \tau^2\sigma = (135)(24).$$

Example 1.3.4 (4.)

$S_3 = \{e, (12), (13), (23), (123), (132)\}$. Then $e^2 = e$ and $|e| = 1$; and

$$(12)^2 = e, \quad \text{so} \quad |(12)| = |(13)| = |(23)| = 2,$$

while $|(123)| = |(132)| = 3$.

Example 1.3.5 (5.)

$(1\ 12\ 8\ 10\ 4)(2\ 13)(5\ 11\ 7)(6\ 9)$ has order

$$\text{lcm}(5, 2, 3, 2) = 30.$$

Theorem 1.3.2 S_Ω is a group

Let Ω be a set and S_Ω be the set of bijections $\Omega \rightarrow \Omega$. Then S_Ω is a group under composition, and S_Ω is infinite.

Proof: We need to prove:

1. S_Ω is a group;
2. S_Ω is infinite.
1. **Closure:** let $\sigma, \tau \in S_\Omega$; σ and τ are bijections from Ω to Ω , so $\sigma\tau$ is also a bijection. Hence $\sigma\tau \in S_\Omega$.

Associativity: inherited (from composition of functions).

Identity: $\varepsilon(n) = n$, $\varepsilon \in S_\Omega$, and $\varepsilon \circ \sigma = \sigma \circ \varepsilon = \sigma$.

Inverse: $\sigma^{-1} : \Omega \rightarrow \Omega$ exists for all $\sigma \in S_\Omega$, since every bijection has an inverse. So $\sigma^{-1} \in S_\Omega$.

2. S_Ω is infinite. For $n \geq 2$, define the transposition $(1\ n)$. Then

$$(1\ 2) \neq (1\ 3) \neq \dots,$$

so they are distinct; continuing this way we get $(1\ 2), (1\ 3), \dots$. So S_Ω is infinite.



Example 1.3.6 (20.)

$$S_3 = \langle r, s \mid r^3 = e, s^2 = e, srs = r^{-1} \rangle$$

by letting $r = (1\ 2\ 3)$ and $s = (1\ 2)$.

1.4 Matrix Groups

Definition 1.4.1: Field

A **field** is a set F together with two binary operators $+$ and $\cdot$ on F such that:

- $(F, +)$ is an abelian group (with identity 0);
- $(F - \{0\}, \cdot)$ is also an abelian group;
- the distributive law

$$a \cdot (b + c) = a \cdot b + a \cdot c$$

holds for all $a, b, c \in F$.

Note:-

For any field F , we write $F^\times = F - \{0\}$.

Definition 1.4.2: $GL_n(F)$

For all $n \in \mathbb{Z}^+$, let

$$GL_n(F) = \{A \mid A \text{ is an } n \times n \text{ matrix with entries from } F \text{ and } \det(A) \neq 0\}.$$

Claim 1.4.1 $GL_n(F)$ is a group

$GL_n(F)$ is a group under matrix multiplication. It is called the **general linear group** of degree n .

Proof:

1. **Closure:** let $A, B \in GL_n(F)$ and let AB be their product (computed with the usual matrix-multiplication rules). Since $\det(AB) = \det(A)\det(B)$, if $\det(A) \neq 0$ and $\det(B) \neq 0$ then $\det(AB) \neq 0$. So $GL_n(F)$ is closed under multiplication.
2. **Associativity:** inherited from matrix multiplication.
3. **Identity:** the identity matrix I satisfies $AI = IA = A$.
4. **Inverse:** $\det(A) \neq 0$ iff A has a matrix inverse. So A^{-1} exists for all $A \in GL_n(F)$, and

$$AA^{-1} = A^{-1}A = I.$$

**1.4.1 Exercises: counting matrices****Example 1.4.1 (1.)**

$|GL_2(F_2)|$: a 2×2 matrix with entries from $F_2 = \{0, 1\}$. How many choices make the matrix invertible (i.e. the two columns c_1, c_2 are linearly independent)?

Writing $A = (c_1 \ c_2)$, for c_1 it can't be $\begin{pmatrix} 0 \\ 0 \end{pmatrix}$, so there are 3 choices for c_1 . Now c_2 cannot have a 0 in the same row as c_1 (and it can't be $\begin{pmatrix} 0 \\ 0 \end{pmatrix}$), so there are $4 - 2 = 2$ choices for c_2 . Hence

$$3 \times 2 = 6 = |GL_2(F_2)|.$$

1.5 The Quaternion Group

Definition 1.5.1: The quaternion group Q_8

Q_8 is defined by

$$Q_8 = \{1, -1, i, -i, j, -j, k, -k\} = \{\pm 1, \pm i, \pm j, \pm k\},$$

where multiplication $\cdot$ is defined as follows: for all $a \in Q_8$,

$$1 \cdot a = a \cdot 1 = a,$$

and for all $a \in Q_8$,

$$(-1) \cdot (-1) = 1, \quad (-1) \cdot a = a \cdot (-1) = -a.$$

The products of i, j, k are:

$$\begin{aligned} i \cdot i &= j \cdot j = k \cdot k = -1, \\ i \cdot j &= k, \quad j \cdot i = -k, \\ j \cdot k &= i, \quad k \cdot j = -i, \\ k \cdot i &= j, \quad i \cdot k = -j. \end{aligned}$$

1.6 Homomorphisms and Isomorphisms

Definition 1.6.1: Homomorphism

Let $(G, *)$ and $(H, \cdot)$ be groups. A map $\varphi : G \rightarrow H$ s.t. for all $x, y \in G$,

$$\varphi(x * y) = \varphi(x) \cdot \varphi(y)$$

is called a **homomorphism**. Or simply $\varphi(xy) = \varphi(x)\varphi(y)$, if the binary operation is not specified.

Definition 1.6.2: Isomorphism

$\varphi : G \rightarrow H$ is called an **isomorphism**, and G and H are said to be isomorphic (written $G \cong H$), if

1. φ is a homomorphism,
2. φ is a bijection.

In MAT 247 we learnt that an isomorphism of vector spaces is a map $T : V \rightarrow W$ which is linear and bijective. And by definition, linear means

$$T(x + y) = T(x) + T(y),$$

which agrees with our previous definition of homomorphism.

1.6.1 Examples of isomorphisms

Example 1.6.1 (1.)

$G \cong G$ for all groups G , since we can choose $\varphi : G \rightarrow G$, $\varphi(x) = x$. Bijection is obvious; check the homomorphism property:

$$\forall x, y \in G, \quad \varphi(xy) = xy = \varphi(x)\varphi(y). \quad \odot$$

Example 1.6.2 (2.)

$\exp : \mathbb{R} \rightarrow \mathbb{R}^+$ defined by $\exp(x) = e^x$ is an isomorphism from $(\mathbb{R}, +)$ to $(\mathbb{R}^+, \times)$. Indeed $\exp$ is a bijection since $\log e$ is its inverse function, and $\exp$ preserves the group operations, since

$$\exp(x + y) = e^{x+y} = e^x e^y = \exp(x) \exp(y).$$

Example 1.6.3 (3.)

Isomorphism type depends only on the cardinality of the underlying set being permuted. Let Δ and Σ be nonempty sets. The symmetric groups S_Δ and S_Σ are isomorphic if and only if $|\Delta| = |\Sigma|$.

Intuitively, if $|\Delta| = |\Sigma|$ there exists a bijection $\theta : \Delta \rightarrow \Sigma$ sending each $x \in \Delta$ to $\theta(x) \in \Sigma$. To obtain $\varphi : S_\Delta \rightarrow S_\Sigma$, let $\sigma \in S_\Delta$ be a permutation of Δ ; then $\varphi(\sigma)$ is the permutation of Σ which moves the elements of Σ in the way θ moves the corresponding elements of Δ : if $\sigma(x) = y$ for some $x, y \in \Delta$, then

$$\varphi(\sigma)(\theta(x)) = \theta(y).$$

So $x \rightarrow y$ is translated to $\theta(x) \rightarrow \theta(y)$. Since θ is invertible, this map between symmetric groups has an inverse.

The intuition is only an idea: for a final proof we must define $\varphi(\sigma)$ precisely, namely $\varphi(\sigma) = \theta \circ \sigma \circ \theta^{-1}$, and then check that it is well defined ($\varphi(\sigma) \in S_\Sigma$) and that it is an isomorphism (a homomorphism and a bijection).

Conversely, if $S_\Delta \cong S_\Sigma$ then $|\Delta| = |\Sigma|$. We prove only the finite case: if $S_\Delta \cong S_\Sigma$, then $|S_\Delta| = |S_\Sigma|$ because there is a bijection between them. If Δ is finite of order n then $|S_\Delta| = n!$, and if Σ has order m then $|S_\Sigma| = m!$. Since $|S_\Delta| = |S_\Sigma|$, we get $n! = m!$, hence $n = m$.

A common use of isomorphism is in *classification theorems*.

Definition 1.6.3: Classification theorem

A **classification theorem** is a theorem that tells you exactly when objects are the same up to isomorphism. Formally: let C be a class of mathematical objects, such as groups, rings, vector spaces, ... A classification theorem gives a list of standard objects $\{M_i : i \in I\}$ s.t.

1. **Existence:** every object $X \in C$ is isomorphic to some M_i ;
2. **Uniqueness:** $M_i \cong M_j$ iff $i = j$.

Equivalently, a classification theorem often gives an *invariant* F s.t. $X \cong Y \iff F(X) = F(Y)$; such an $F(\cdot)$ is called a **complete invariant**.

Example 1.6.4 (Vector spaces)

In linear algebra, $V \cong W \iff \dim V = \dim W$ for finite-dimensional vector spaces over some field.

For example, a classification theorem: any non-abelian group of order 6 is isomorphic to S_3 . By this theorem we can obtain that $D_6 \cong S_3$ and $GL_2(F_2) \cong S_3$.

The textbook notes that it is difficult to determine whether or not two groups are isomorphic: constructing a specific map that preserves the group operation, or proving that no such map exists, is computationally unfeasible.

However, to *disprove* isomorphism is sometimes easy. For example, if $\varphi : G \rightarrow H$ is an isomorphism, then:

- (a) $|G| = |H|$;
- (b) G is abelian $\iff H$ is abelian;
- (c) $\forall x \in G, |x| = |\varphi(x)|$.

Example 1.6.5 ($S_3 \not\cong \mathbb{Z}/6\mathbb{Z}$)

S_3 is not isomorphic with $\mathbb{Z}/6\mathbb{Z}$, since S_3 is not abelian. Let $\sigma = (1\ 2)$, $\tau = (2\ 3)$. Then

$$\sigma\tau = (1\ 2)(2\ 3) = (1\ 2\ 3),$$

while

$$\tau\sigma = (2\ 3)(1\ 2) = (1\ 3\ 2),$$

and $\mathbb{Z}/6\mathbb{Z}$ is abelian: for all $a, b \in \mathbb{Z}/6\mathbb{Z}$, $a + b \equiv b + a \pmod{6}$.

Theorem 1.6.1 Homomorphism from generators

Let G be a finite group of order n , and let $S = \{s_1, \dots, s_m\}$ be generators. Let H be another group and $\{r_1, \dots, r_m\}$ be elements of H . Suppose that any relation satisfied in G by s_i is also satisfied in H when s_i is replaced by r_i . Then there is a unique homomorphism $\varphi : G \rightarrow H$ which maps $s_i \mapsto r_i$.

If G has a presentation, then one only needs to check the relations specified by the presentation. If H is generated by $r_1, \dots, r_m$, then φ is surjective. If H has the same order as G , then φ is injective, hence φ is an isomorphism and $H \cong G$.

In words: when we know the generators of a group G , then for H we can find elements $r_i \in H$ which satisfy the corresponding relations. Then the map $\varphi : G \rightarrow H$ sending the generators correspondingly is a homomorphism; if the elements generate H it is surjective, and if $|G| = |H|$ it is injective, hence $G \cong H$.

For vector spaces this is the familiar statement that a linear transformation $V \rightarrow W$ is

determined by where a basis of V is sent: if $\dim V = \dim W$ and the images span W , the map is invertible.

Example 1.6.6 (Application of the Fact)

Recall that $D_{2n} = \langle r, s \mid r^n = s^2 = 1, \, srs = r^{-1} \rangle$. Suppose H is a group, and

$$H = \langle a, b \mid a^n = b^2 = 1, \, ba = a^{-1}b \rangle.$$

Define $\varphi : D_{2n} \rightarrow H$ where, in every word, $r \mapsto a$ and $s \mapsto b$.

A common misconception is to think only that $\varphi(r) = a$ and $\varphi(s) = b$. This is true, but it is *not* a homomorphism since $\varphi(rs)$ is not defined. By saying “every word maps”, it means that $\varphi(rs^2r^{-1}) = ab^2a^{-1}$ and so on. This is well defined since the relations are exactly the same; therefore this is a homomorphism.

Moreover, consider D_{2k} where k is an integer dividing n ; write $n = km$ and let

$$D_{2k} = \langle r_1, s_1 \mid r_1^k = s_1^2 = 1, \, s_1r_1 = r_1^{-1}s_1 \rangle.$$

Since $r_1^k = 1$, we have $r_1^n = (r_1^k)^m = 1$, so the relations satisfied by D_{2k} are satisfied by D_{2n} . Therefore $r \mapsto r_1, s \mapsto s_1$ is a homomorphic map. Since $\{r_1, s_1\}$ generates D_{2k} , the map is also surjective. If $n = k$, then $|D_{2n}| = |D_{2k}|$, so φ is injective and they are isomorphic.

Example 1.6.7 (2.)

Consider D_6 and S_3 . Let $a = (1\,2\,3)$ and $b = (1\,2)$. Since

$$D_6 = \langle r, s \mid r^3 = s^2 = 1, \, sr = r^{-1}s \rangle,$$

a, b satisfy the relationships correspondingly:

$$a^3 = (123)(123)(123) = 1 \quad (\text{since } a \text{ is a 3-cycle}), \quad b^2 = (12)(12) = 1 \quad (b \text{ is a 2-cycle}).$$

Moreover,

$$ba = (12)(123) = (23), \quad a^{-1}b = (321)(12) = (23) = ba.$$

So the map $r \mapsto a, s \mapsto b$ is homomorphic. Moreover, since

$$S_3 = \{e, (12), (23), (13), (123), (132)\},$$

we can express them all in a, b , so a, b generate S_3 :

$$e = a^3, \quad (12) = b, \quad (23) = ba, \quad (13) = ab, \quad (123) = a, \quad (132) = a^2.$$

So this is surjective. Since $|S_3| = |D_6|$, φ is injective, hence an isomorphism.

1.6.2 Properties of homomorphisms

Example 1.6.8 (1.)

(a) Since φ is a homomorphism, $\varphi(xx) = \varphi(x)\varphi(x)$, $\varphi(xxx) = \varphi(xx)\varphi(x) = \varphi(x)\varphi(x)\varphi(x)$. So

$$\varphi(x^n) = \varphi(x)\varphi(x^{n-1}) = \cdots = \varphi(x)^n.$$

(A formal proof requires induction.)

(b) Since $xx^{-1} = 1$, then $\varphi(xx^{-1}) = \varphi(x)\varphi(x^{-1}) = \varphi(1)$. But $\varphi(1) = \varphi(1 \cdot 1) = \varphi(1)\varphi(1)$, so $\varphi(1) = 1$. Thus $\varphi(x)\varphi(x^{-1}) = 1$, so $\varphi(x^{-1}) = \varphi(x)^{-1}$.

Example 1.6.9 (Let $n < 0$, and write $n = -m$ with $m > 0$. Then

$$\begin{aligned}\varphi(x^n) &= \varphi(x^{-m}) = \varphi((x^{-1})^m) \stackrel{(1a)}{=} \varphi(x^{-1})^m \\ &= (\varphi(x)^{-1})^m = \varphi(x)^{-m} = \varphi(x)^n.\end{aligned}$$

)

Example 1.6.10 (2.)

Assume $x^n = 1$ (let m be the smallest positive exponent such that $x^m = 1$, i.e. $m = |x|$). Then

$$\varphi(x^n) = \varphi(1) = 1 = \varphi(x)^n.$$

Hence $|\varphi(x)| \mid |x|$. Let $k = |\varphi(x)|$; then $\varphi(x)^k = 1$.

Example 1.6.11 (If, moreover, φ is injective, then $\varphi(x^k) = 1 = \varphi(1) \Rightarrow x^k = 1$; by minimality of $m = |x|$, $m \mid k$, so $m = k$. Hence $|\varphi(x)| = |x|$.)

If x has infinite order but $\varphi(x)$ has finite order k , then

$$\varphi(x^k) = \varphi(x)^k = 1 = \varphi(1).$$

Injectivity implies $x^k = 1$, a contradiction. So if φ is injective and $|x| = \infty$, then $|\varphi(x)| = \infty$.

For $n \in \mathbb{Z}^+$, define

$$A = \{x \in G \mid |x| = n\}, \quad B = \{y \in H \mid |y| = n\}.$$

Since φ preserves order, it maps A into B . Because φ is bijective, the restriction $\varphi|_A : A \rightarrow B$ is bijective, so $|A| = |B|$.

Example 1.6.12 (If φ is only a homomorphism (not injective), this need not hold. For instance $\varphi : D_6 \rightarrow \{e\}$ is a homomorphism with $\varphi(g) = e$ for all g , so $|\varphi(g)| = 1$ for every

g , whereas $|g|$ varies.)

Example 1.6.13 (3.)

G is abelian $\iff H$ is abelian.

($\Rightarrow$) If G is abelian, then for any $g_1, g_2 \in G$, $g_1g_2 = g_2g_1$. Then

$$\varphi(g_1g_2) = \varphi(g_2g_1) = \varphi(g_1)\varphi(g_2) = \varphi(g_2)\varphi(g_1).$$

Since $\forall h \in H$, $\exists g \in G$ with $\varphi(g) = h$ (surjectivity), for all $h_1, h_2 \in H$ there are $g_1, g_2 \in G$ s.t. $h_1 = \varphi(g_1)$, $h_2 = \varphi(g_2)$. Then

$$h_1h_2 = \varphi(g_1)\varphi(g_2) = \varphi(g_2g_1) = h_2h_1.$$

Hence H is abelian.

($\Leftarrow$) If H is abelian, then for any $h_1, h_2 \in H$, $h_1h_2 = h_2h_1$. By surjectivity there exist $g_1, g_2 \in G$ s.t. $\varphi(g_1) = h_1$ and $\varphi(g_2) = h_2$. So

$$h_1h_2 = \varphi(g_1)\varphi(g_2) = h_2h_1 = \varphi(g_2)\varphi(g_1).$$

Thus $\varphi(g_1)\varphi(g_2) = \varphi(g_2)\varphi(g_1) = \varphi(g_2g_1)$. By injectivity $\varphi(g_1g_2) = \varphi(g_2g_1)$, hence $g_1g_2 = g_2g_1$. Since φ is bijective and h_1, h_2 were arbitrary, $\forall g_1, g_2 \in G$, $g_1g_2 = g_2g_1$, so G is abelian.

Note:-

To ensure “ G abelian $\Rightarrow H$ abelian”, we only need φ to be surjective (as in the proof above).

1.6.3 Groups that are not isomorphic

Example 1.6.14 (4.)

In $\mathbb{C}^\times$, $i^4 = 1$, so $|i| = 4$. Suppose $x \in \mathbb{R}^\times$ has order 4; then $x^4 = 1$, so $x^4 - 1 = 0$, i.e. $(x^2 + 1)(x^2 - 1) = 0$. Then $x = \pm 1$, whose orders are 1 and 2. So $\mathbb{R}^\times$ has no element of order 4. Since order is not preserved, $\mathbb{R}^\times$ and $\mathbb{C}^\times$ cannot be isomorphic.

Example 1.6.15 (5.)

$\mathbb{Q}$ is countable, $\mathbb{R}$ is not countable. Since $|\mathbb{Q}| \neq |\mathbb{R}|$, there is no bijection $\varphi : \mathbb{Q} \rightarrow \mathbb{R}$, so $(\mathbb{Q}, +)$ and $(\mathbb{R}, +)$ are not isomorphic.

Example 1.6.16 (6.)

Suppose, for contradiction, that there exists an isomorphism $\varphi : (\mathbb{Z}, +) \rightarrow (\mathbb{Q}, +)$. By

surjectivity there is $n \in \mathbb{Z}$ with $\varphi(n) = 1$, and there is $m \in \mathbb{Z}$ with $\varphi(m) = \frac{1}{2}$. Then

$$\varphi(2m) = 2\varphi(m) = 2 \cdot \frac{1}{2} = 1 = \varphi(n),$$

so by injectivity $n = 2m$. Now since $\varphi(1) \in \mathbb{Q}$, there is $q \in \mathbb{Q}$ with $2q = \varphi(1)$, and by surjectivity $\varphi(m) = q$ for some $m \in \mathbb{Z}$. Then

$$\varphi(2m) = 2\varphi(m) = 2q = \varphi(1),$$

so $2m = 1$, a contradiction. Hence $(\mathbb{Z}, +) \not\cong (\mathbb{Q}, +)$.

1.6.4 Constructing isomorphisms

Example 1.6.17 (10.)

Let $\theta : \Delta \rightarrow \Sigma$ be a bijection and define $\varphi : S_\Delta \rightarrow S_\Sigma$ by $\varphi(\sigma) = \theta \circ \sigma \circ \theta^{-1}$.

- (a) θ is a bijection, so $\theta^{-1} : \Sigma \rightarrow \Delta$ is a bijection. Hence the composition $\theta \circ \sigma \circ \theta^{-1} : \Sigma \rightarrow \Sigma$ is a bijection from Σ to itself, so $\varphi(\sigma) \in S_\Sigma$; it is well defined.
- (b) φ is a bijection from S_Δ to S_Σ . Let $\psi = \theta^{-1} \circ \rho \circ \theta$ be the inverse of φ (well defined as in (a)). Then

$$\varphi \circ \psi(\rho) = \theta \circ (\theta^{-1} \circ \rho \circ \theta) \circ \theta^{-1} = \rho,$$

and $\psi \circ \varphi(\sigma) = \sigma$. So $\psi = \varphi^{-1}$, i.e. φ is a bijection.

- (c) Need to show $\varphi(\sigma \circ \tau) = \varphi(\sigma) \circ \varphi(\tau)$:

$$\varphi(\sigma \circ \tau) = \theta \circ (\sigma \circ \tau) \circ \theta^{-1} = (\theta \circ \sigma \circ \theta^{-1}) \circ (\theta \circ \tau \circ \theta^{-1}) = \varphi(\sigma) \circ \varphi(\tau).$$

Hence φ is an isomorphism and $S_\Delta \cong S_\Sigma$.

Note:-

So $S_\Delta \cong S_\Sigma$ if $|\Delta| = |\Sigma|$.

Example 1.6.18 (11.)

Assume $\varphi : A \times B \rightarrow B \times A$, $\varphi(a, b) = (b, a)$. It is a homomorphism:

$$\varphi((a, b)(c, d)) = \varphi(ac, bd) = (bd, ac), \quad \varphi(a, b)\varphi(c, d) = (b, a)(d, c) = (bd, ac).$$

It is bijective: let $\theta : B \times A \rightarrow A \times B$ be $\theta(b, a) = (a, b)$. Then $\theta \circ \varphi = \text{id}_{A \times B}$ and $\varphi \circ \theta = \text{id}_{B \times A}$, so $\theta = \varphi^{-1}$, i.e. φ is a bijection. Hence $A \times B \cong B \times A$.

Example 1.6.19 (12.)

Define $\varphi : (G \times B) \times C \rightarrow G \times (B \times C)$ by $\varphi((a, b), c) = (a, (b, c))$. Check the homomorphism

property:

$$\begin{aligned}\varphi(((a, b), c) \cdot ((d, e), f)) &= \varphi((a, b)(d, e), cf) = \varphi((ad, be), cf) = (ad, (be, cf)), \\ \varphi((a, b), c) \cdot \varphi((d, e), f) &= (a, (b, c)) \cdot (d, (e, f)) = (ad, (be, cf)).\end{aligned}$$

These agree, so φ is an isomorphism — associativity of the direct product. It is a bijection: the map $\Phi : A \times H \rightarrow G \times C$ given by $\Phi(a, (b, c)) = ((a, b), c)$ satisfies $\Phi = \varphi^{-1}$; hence φ is a bijection and an isomorphism.

Example 1.6.20 (13.)

Recall the notion of subgroup.

Definition 1.6.4: Subgroup

Assume H is a nonempty subset of $(G, *)$ which is closed under the binary operation of G and closed under inverses. Then H is called a **subgroup** of G .

Proposition 1.6.1 Subgroup is a group

Moreover, H is a group under the operation $*$ restricted to H .

Proof: Let $H \subseteq (G, *)$ be nonempty and closed under $*$ and inverse. Let $h \in H$; then $h^{-1} \in H$, so $h \cdot h^{-1} = 1 \in H$. Hence H has an identity. Associativity is inherited from G . Thus $(H, *)$ is a group. ☺

Theorem 1.6.2 $\varphi(G)$ is a subgroup

Back to question 13: to show $\varphi(G)$ is a subgroup, we prove that it is closed under $*$ and inverse, and is nonempty.

Proof: **Closure under $*$:** for all $\varphi(a), \varphi(b) \in \varphi(G)$,

$$\varphi(a)\varphi(b) = \varphi(ab) \in \varphi(G)$$

(because φ is a homomorphism).

Closure under inverse: for all $\varphi(a) \in \varphi(G)$,

$$\varphi(a)^{-1} = \varphi(a^{-1}) \in \varphi(G).$$

Nonempty: $e_H \in \varphi(G)$ since $\varphi(e_G) = e_H$.

☺

Since $\varphi(G) \neq \emptyset$, $\varphi(G)$ is a subgroup of H . If φ is injective, then $\Phi : G \rightarrow \varphi(G)$ given by $\Phi(g) = \varphi(g)$ is a homomorphism; it is surjective since every element of $\varphi(G)$ is $\varphi(g)$ for some g , and it is injective because φ is. Hence $G \cong \varphi(G)$.

Definition 1.6.5: Kernel

Let G, H be groups and $\varphi : G \rightarrow H$ be a homomorphism. Define the **kernel** of φ to be

$$\ker \varphi = \{g \in G \mid \varphi(g) = 1_H\}.$$

Theorem 1.6.3 Kernel is a subgroup

$\ker \varphi$ is a subgroup of G .

Proof: To prove it's a subgroup, prove closure under the binary operation and inverse. Since $\varphi(1_G) = 1_H$, $1_G \in \ker \varphi$, so $\ker \varphi$ is not empty. For all $a, b \in \ker \varphi$: $\varphi(a) = 1_H$ and $\varphi(b) = 1_H$, so

$$\varphi(ab) = \varphi(a)\varphi(b) = 1_H \cdot 1_H = 1_H,$$

and $ab \in \ker \varphi$. For all $g \in \ker \varphi$:

$$\varphi(g^{-1}) = \varphi(g)^{-1} = (1_H)^{-1} = 1_H,$$

so $g^{-1} \in \ker \varphi$. Thus $\ker \varphi$ is a subgroup of G . ☺

We also have $\ker \varphi = \{1_G\}$ iff φ is injective. Indeed, if φ is injective then $g \in \ker \varphi$ gives $\varphi(g) = 1_H = \varphi(1_G)$, so $\ker \varphi = \{1_G\}$. Conversely, if $\ker \varphi = \{1_G\}$ and $\varphi(g_1) = \varphi(g_2)$, then $\varphi(g_1g_2^{-1}) = 1_H$, so $g_1g_2^{-1} \in \ker \varphi = \{1_G\}$, hence $g_1 = g_2$.

Definition 1.6.6: Automorphism

Let G be a group, and let $\text{Aut}(G)$ be the set of all isomorphisms from G to G . $\text{Aut}(G)$ is the **automorphism group** of G , and its elements are the automorphisms of G .

Note:-

That $\text{Aut}(G)$ is a group is standard.

Example 1.6.21 (23.)

Let G be a finite group which possesses an automorphism σ s.t. $\sigma(g) = g$ iff $g = 1$. If σ^2 is the identity map from G to G , prove that G is abelian (such an automorphism σ is called fixed-point-free of order 2).

Proof: Define the map $f : G \rightarrow G$ by $f(x) = x^{-1}\sigma(x)$. For all $x \in G$,

$$f(\sigma(x)) = \sigma(x)^{-1}\sigma(\sigma(x)) = \sigma(x)^{-1}x.$$

Since σ is fixed-point-free, f is injective. Since G is finite, f is injective, hence bijective; so f is surjective, i.e. any $g \in G$ can be written as $f(x)$ for some x .

So every $g \in G$ can be written in the form $x^{-1}\sigma(x)$ for some $x \in G$. Let $g \in G$ be arbitrary; there is $x \in G$ with $g = x^{-1}\sigma(x)$. Then

$$\sigma(g) = \sigma(x^{-1}\sigma(x)) = \sigma(x^{-1}) \cdot \sigma^2(x) = \sigma(x)^{-1} \cdot x = (x^{-1}\sigma(x))^{-1} = g^{-1}.$$

Thus $\sigma(g) = g^{-1}$ for all g . Then for any $g, h \in G$,

$$\sigma(gh) = \sigma(g)\sigma(h) = g^{-1}h^{-1},$$

but also

$$\sigma(gh) = (gh)^{-1} = h^{-1}g^{-1} = \sigma(h)\sigma(g).$$

So $\sigma(g)\sigma(h) = \sigma(h)\sigma(g)$. Since σ is surjective, $gh = hg$, and G is abelian. ☺

Definition 1.6.7: Fixed-point-free

A map $\sigma : G \rightarrow G$ is called fixed-point-free when the only element it leaves unmoved is the identity:

$$\sigma(g) = g \iff g = 1.$$

1.7 Group Action

Definition 1.7.1: Group action

A **group action** of a group G on a set A is a map from $G \times A$ to A , written $g \cdot a$, for all $g \in G$ and $a \in A$, satisfying the following properties:

1. $g_1 \cdot (g_2 \cdot a) = (g_1 g_2) \cdot a$;
2. $1 \cdot a = a$ for all $a \in A$.

Proposition 1.7.1 Group action gives permutations

Let a group G act on a set A . For each fixed $g \in G$, we get a map σ_g defined by

$$\sigma_g : A \rightarrow A, \quad \sigma_g(a) = g \cdot a.$$

We prove two important facts:

1. For each fixed $g \in G$, σ_g is a permutation of A .
2. The map from G to S_A defined by $g \mapsto \sigma_g$ is a homomorphism.

Proof: (1) For all $a \in A$,

$$(\sigma_{g^{-1}} \circ \sigma_g)(a) = \sigma_{g^{-1}}(\sigma_g(a)) = \sigma_{g^{-1}}(g \cdot a) = g^{-1} \cdot (g \cdot a) = (g^{-1}g) \cdot a = 1 \cdot a = a.$$

So $\sigma_{g^{-1}} \circ \sigma_g$ is the identity map from A to A ; similarly for $\sigma_g \circ \sigma_{g^{-1}}$. Thus σ_g has a two-sided inverse, hence σ_g is a permutation of A .

(2) Let $\varphi : G \rightarrow S_A$ be $\varphi(g) = \sigma_g$. By (1), $\sigma_g \in S_A$. To see φ is a homomorphism, we need to show $\varphi(g_1 g_2) = \varphi(g_1) \varphi(g_2)$. For all $a \in A$,

$$\varphi(g_1 g_2)(a) = \sigma_{g_1 g_2}(a) = (g_1 g_2) \cdot a = g_1 \cdot (g_2 \cdot a) = \sigma_{g_1}(\sigma_{g_2}(a)) = (\varphi(g_1) \circ \varphi(g_2))(a).$$

☺

Note:-

A group action G on A intuitively means that every element $g \in G$ acts as a permutation on A in a way consistent with the group operation of G .

Definition 1.7.2: Permutation representation

The homomorphism $\varphi : G \rightarrow S_A$ is called the **permutation representation** associated to the given action.

Note:-

If the process is reversible in the sense that $\varphi : G \rightarrow S_A$, then the map from $G \times A$ to A defined by

$$g \cdot a = \varphi(g)(a), \quad \forall g \in G, a \in A,$$

satisfies the properties of a group action of G on A .

Chapter 2

Subgroups

2.1 Subgroup definition and basic examples

Definition 2.1.1: Subgroup

For a group G , a set $H \subseteq G$ is a **subgroup** if H is nonempty and closed under the binary operation and inverse.

Definition 2.1.2: Trivial subgroup

The trivial subgroup H of a group G is $H = \{1\}$.

Proposition 2.1.1 Subgroup criterion

A subset H of G is a subgroup of G iff

1. $H \neq \emptyset$;
2. $\forall x, y \in H, xy^{-1} \in H$.

If G is finite, the criterion simplifies: it is enough to check that $H \neq \emptyset$ and is closed under multiplication. Indeed, for a finite group, let $x \in H$; the powers $x, x^2, x^3, \dots$ all lie in H , and since H is finite some two coincide, say $x^i = x^j$ with $i > j$. Then $e = x^{i-j}$, and $x^{-1} = x^{i-j-1} \in H$. Since x was arbitrary, every element of H has its inverse in H .

2.1.1 Verifying subgroups

Example 2.1.1 (1.)

Let G be the group and H be the subset in question. Use the subgroup criterion to show $H \subseteq G$.

- (a) $0 + 0 \in H$, so H is nonempty; and

$$(a + ai) - (b + bi) = (a - b) + (a - b)i \in H,$$

so $H \subseteq G$.

(b) $|1| = 1$, so $H \neq \emptyset$. For $z, w \in H$, $|z| = |w| = 1$; since $w \neq 0$ the inverse exists. Then

$$|zw^{-1}| = |z| |w^{-1}| = \frac{|z|}{|w|} = 1,$$

so $zw^{-1} \in H$, and $H \subseteq G$.

(c) Fix $n \in \mathbb{Z}^+$; $0 = \frac{0}{k}$ for some $k \neq 0$ with $k \mid n$, so $0 \in G$ and H is nonempty. Let $\frac{a}{b}, \frac{c}{d} \in H$ where $b \mid n$ and $d \mid n$. Then

$$\frac{a}{b} - \frac{c}{d} = \frac{ad - bc}{bd},$$

and $bd \mid n$. So $H \subseteq G$.

(d) Fix n ; $0 = \frac{0}{1}$ and $\gcd(n, 1) = 1$, so $0 \in H$. Let $\frac{a}{b}, \frac{c}{d} \in H$; then $\gcd(b, n) = 1$ and $\gcd(d, n) = 1$, so

$$\frac{a}{b} - \frac{c}{d} = \frac{ad - bc}{bd}.$$

If $\gcd(bd, n) \neq 1$, then $\exists m \in \mathbb{Z}^+$ with $m \mid bd$ and $m \mid n$; then $m \mid b$ or $m \mid d$, so $\gcd(b, n) \geq m$ or $\gcd(d, n) \geq m$, a contradiction. Hence $\gcd(bd, n) = 1$, and $H \subseteq G$.

(e) $1^2 = 1$, so $H \neq \emptyset$. Let $x, y \in H$, then $x^2, y^2 \in \mathbb{Q}$. Now let $x = \frac{a}{b}$, $y = \frac{c}{d}$ with $a, b, c, d \in \mathbb{Z}$. Then

$$(xy^{-1})^2 = \frac{x^2}{y^2} = \frac{a^2/b^2}{c^2/d^2} = \frac{a^2 d^2}{b^2 c^2} = \left(\frac{ad}{bc} \right)^2 \in \mathbb{Q},$$

so $xy^{-1} \in H$, and $H \subseteq G$.

Note:-

In part c, the claim “ $bd \mid n$ ” is slightly too strong: bd need not divide n . What matters is that after reducing, the denominator of $\frac{ad-bc}{bd}$ divides $\text{lcm}(b, d)$, hence n . So the conclusion still holds.

2.1.2 The torsion subgroup

Example 2.1.2 (Let G be an abelian group. Prove that $\{g \in G \mid |g| < \infty\}$ is a subgroup of G .)

Definition 2.1.3: Torsion group

The set

$$\text{Tor}(G) = \{g \in G \mid |g| < \infty\}$$

is called the **torsion group** of G .

Proposition 2.1.2 Torsion group is a subgroup

$\text{Tor}(G)$ is a subgroup of G .

Proof: Since $|e_G| = 1 < \infty$, $e_G \in \text{Tor}(G)$, so $\text{Tor}(G) \neq \emptyset$. For $x, y \in \text{Tor}(G)$, let $|x| = m$ and $|y| = n$, with $m, n < \infty$. Then, since G is abelian,

$$(xy^{-1})^{mn} = x^{mn}y^{-mn} = 1,$$

so $|xy^{-1}| \leq mn < \infty$ and $xy^{-1} \in \text{Tor}(G)$. Hence $\text{Tor}(G) \subseteq G$, i.e. $\text{Tor}(G)$ is a subgroup. $\odot$

Example 2.1.3 (Non-abelian counterexample)

When G is non-abelian, $\text{Tor}(G)$ need not be a subgroup. Take the infinite dihedral group $D_\infty = \langle r, s \mid s^2 = e, srs = r^{-1} \rangle$. Here $|s| = 2$, and $|rs| = 2$ since

$$(rs)^2 = rsrs = r(srs) = rr^{-1} = 1.$$

But $s^{-1}(sr) = s(sr) = r$ and $|r| = \infty$, so $s^{-1}(sr) \notin \text{Tor}(D_\infty)$; the set of finite-order elements is not closed under multiplication.

Example 2.1.4 ($GL_2(\mathbb{R})$)

Similarly, $GL_2(\mathbb{R})$ is not abelian. For

$$A = \begin{pmatrix} -1 & 0 \\ 0 & 1 \end{pmatrix}, \quad B = \begin{pmatrix} -1 & 1 \\ 0 & 1 \end{pmatrix},$$

we have $|A| = |B| = 2$, and

$$AB = \begin{pmatrix} 1 & -1 \\ 0 & 1 \end{pmatrix}.$$

By induction one shows $(AB)^n = \begin{pmatrix} 1 & -n \\ 0 & 1 \end{pmatrix}$ for all $n \in \mathbb{Z}^+$, so $|AB| = \infty$ and $AB \notin \text{Tor}(GL_2(\mathbb{R}))$.

2.1.3 Subgroups of matrix groups and intersections**Example 2.1.5** (9.)

Let $G = GL_2(F)$, where F is any field.

Definition 2.1.4: Special linear group

$$SL_n(F) = \{A \in GL_n(F) \mid \det(A) = 1\}.$$

$SL_n(F)$ is called the **special linear group**.

Proposition 2.1.3 $SL_n(F) \leq GL_n(F)$

$SL_n(F) \subseteq GL_n(F)$ is a subgroup.

Proof: Since $\det(I_n) = 1$, $SL_n(F) \neq \emptyset$. Let $A, B \in SL_n(F)$. Then

$$\det(AB^{-1}) = \det(A) \det(B^{-1}) = \frac{\det(A)}{\det(B)} = 1,$$

so $AB^{-1} \in SL_n(F)$. Hence $SL_n(F) \subseteq GL_n(F)$. ☺

Example 2.1.6 (10.a))

Let $H \subseteq G$ and $K \subseteq G$. Then $e \in H \cap K$, so $H \cap K \neq \emptyset$. Let $a, b \in H \cap K$; then $a, b \in H$ and $a, b \in K$, hence $ab^{-1} \in H$ and $ab^{-1} \in K$, so $ab^{-1} \in H \cap K$.

Example 2.1.7 (10.b))

Let I be an indexing set. Consider the collection of subgroups $\{H_i\}_{i \in I}$ of G , and let

$$H = \bigcap_{i \in I} H_i.$$

Since $e \in H_i$ for all $i \in I$, we have $e \in H$. Suppose $a, b \in H$; then for all $i \in I$, $a, b \in H_i$. Since each H_i is a subgroup of G , $ab^{-1} \in H_i$ for all $i \in I$, so $ab^{-1} \in H$. Hence $H \subseteq G$, i.e. H is a subgroup.

2.2 Centralizers and normalizers, stabilizers and kernels

Definition 2.2.1: Centralizer

Define

$$C_G(A) = \{g \in G \mid gag^{-1} = a \ \forall a \in A\}.$$

This subset of G is called the **centralizer** of A in G . Since $gag^{-1} = a$ iff $ga = ag$, $C_G(A)$ is the set of elements of G which commute with every element of A .

Proposition 2.2.1 $C_G(A)$ is a subgroup

$C_G(A)$ is a subgroup of G .

Proof: $C_G(A) \neq \emptyset$ since $e \in C_G(A)$ (the definition of identity gives $ea = ae$ for all $a \in A$). Assume $x, y \in C_G(A)$: for all $a \in A$, $xax^{-1} = a$ and $yay^{-1} = a$, so $a = y^{-1}ay$, and $y^{-1} \in C_G(A)$ (closed under inverse). Finally,

$$(xy)a(xy)^{-1} = (xy)a(y^{-1}x^{-1}) = x(yay^{-1})x^{-1} = xax^{-1} = a,$$

so $xy \in C_G(A)$. ☺

Note:-

So $xy \in C_G(A)$; closed under products, $C_G(A) \subseteq G$.

Definition 2.2.2: Center

$$Z(G) = \{g \in G \mid gx = xg \ \forall x \in G\}.$$

This subset of G is the **center** of G .

Note:-

Then by definition, for all abelian groups their center is the whole group. Also

$$Z(G) = C_G(G),$$

so $Z(G) \subseteq G$.

Definition 2.2.3: Normalizer

Let

$$gAg^{-1} = \{gag^{-1} \mid a \in A\}.$$

The **normalizer** of A in G is

$$N_G(A) = \{g \in G \mid gAg^{-1} = A\}.$$

In other words, the set of elements g in G such that conjugating A by g gives A back.

Note:-

Note that gAg^{-1} allows the items in A to shuffle: $ga_1g^{-1} = a_2$ also works if $a_1, a_2 \in A$. So only when $\forall x \in A, gxg^{-1} = x$ does g fix each element of A , but the converse need not hold. Hence $C_G(A) \subseteq N_G(A)$.

Example 2.2.1 (1.)

Let $G = D_8$ and let $A = \{1, r, r^2, r^3\}$. Then $C_{D_8}(A) = A$.

Proof: Since r commutes with all of its powers, $A \subseteq C_{D_8}(A)$. The elements of D_8 not in A are all of the form sr^i for some $i \in \{0, 1, 2, 3\}$. If $sr^i \in C_{D_8}(A)$, then since $C_{D_8}(A)$ is a subgroup containing r , we get $s = (sr^i)r^{-i} \in C_{D_8}(A)$, a contradiction. This shows $C_{D_8}(A) = A$. ☺

Example 2.2.2 (2.)

Keep G, A as defined in 1). Then $N_{D_8}(A) = D_8$.

Proof: Since in general the centralizer of a subset is contained in its normalizer, $A \subseteq N_{D_8}(A)$. Now

$$sAs^{-1} = \{s1s^{-1}, srs^{-1}, sr^2s^{-1}, sr^3s^{-1}\} = \{1, r^3, r^2, r\} = A,$$

so $s \in N_{D_8}(A)$. Note that the elements of A are in a different order here, so s is not in the centralizer of A but only in the normalizer. Both r and s belong to $N_{D_8}(A)$; since $\langle r, s \rangle = D_8$, we get $D_8 = N_{D_8}(A)$. ☺

Example 2.2.3 (4.)

The center of D_8 is the subgroup $\langle 1, r^2 \rangle$.

Proof: First, the center of G is contained in $C_G(A)$; by 2), $Z(D_8) \subseteq C_{D_8}(A) = A = \{1, r, r^2, r^3\}$. Since $sr = r^{-1}s \neq rs$, and similarly for r^3 , we have $r, r^3 \notin Z(D_8)$. So $Z(D_8) \subseteq \{1, r^2\}$. Conversely, r commutes with r^2 and s commutes with r^2 ; since r, s generate D_8 , every element of D_8 commutes with r^2 and 1. So $\{1, r^2\} \subseteq Z(D_8)$, hence $\langle 1, r^2 \rangle = Z(D_8)$. ☺

Example 2.2.4 (3.)

Let $G = S_3$ and let A be the subgroup $\{1, (12)\}$. Show that $C_{S_3}(A) = N_{S_3}(A) = A$.

Proof: Since $(12)^2 = 1$, A has two elements that commute with each other, and every element of A commutes with any power of itself. Thus $A \subseteq C_{S_3}(A)$ (and $A \subseteq N_{S_3}(A)$). By Lagrange's theorem $|A| \mid |C_{S_3}(A)|$, i.e. $2 \mid |C_{S_3}(A)|$. Since $C_{S_3}(A) \leq S_3$, $|C_{S_3}(A)| \mid |S_3| = 6$, so $|C_{S_3}(A)| = 2$ or 6. If the order were 6, then $C_{S_3}(A) = S_3$, which is a contradiction, since (12) does not commute with (123) :

$$(12)(123) = (23), \quad (123)(12) = (13).$$

Hence $|C_{S_3}(A)| = 2$, and since $A \subseteq C_{S_3}(A)$ with $|A| = 2$, we get $A = C_{S_3}(A)$.

Now for the normalizer: $sAs^{-1} = \{s1s^{-1}, s(12)s^{-1}\}$. If $s(12)s^{-1} = (12)$, then $s(12) = (12)s$, i.e. s and (12) commute, so $s \in C_{S_3}(A)$. Thus $N_{S_3}(A) \subseteq C_{S_3}(A) = A$. For the reverse inclusion, $\forall a \in A$, $aga^{-1} \in A$ by subgroup closure. Therefore $A = C_{S_3}(A) = N_{S_3}(A)$. ☺

Note:-

The center of S_3 is 1, since $Z(S_3) \subseteq C_{S_3}(A) = A$ and $(12) \notin Z(S_3)$.

2.2.1 Stabilizers and kernels of group actions

Definition 2.2.4: Stabilizer

Let G be a group acting on a set S , and let s be some fixed element of S . The **stabilizer** of s in G is the set

$$G_s = \{g \in G \mid g \cdot s = s\}.$$

Proposition 2.2.2 G_s is a subgroup

$$G_s \leq G.$$

Proof: (1) Let $1 \in G$; $1 \cdot s = s$ for all $s \in S$, so $1 \in G_s$. (2) Let $x, y \in G_s$. Then

$$s = 1 \cdot s = (y^{-1}y) \cdot s = y^{-1} \cdot (y \cdot s) = y^{-1} \cdot s,$$

so $y^{-1} \in G_s$. And

$$(xy) \cdot s = x \cdot (y \cdot s) = x \cdot s = s,$$

so $xy \in G_s$. Hence $G_s \leq G$. ☺

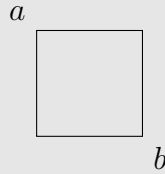
Example 2.2.5 (1.)

The group $G = D_8$ acts on A , the four vertices of a square. The stabilizer G_a of any vertex $a \in A$ is the subgroup of order 2 consisting of 1 and the reflection about the line of symmetry passing through vertex a and the center of the square.

The handwritten note writes this stabilizer as $\{1, r^2\}$; but r^2 is the 180° rotation, which moves every vertex. The stabilizer of a vertex is in fact generated by the reflection through the diagonal through that vertex.

Note:-

Illustration:



Here A is a special element of D_8 : $A = sr^i$, $i \in \{0, 1, 2, 3\}$. It is complicated; we write it as A (a general element of D_8) instead of sr^i , to prevent confusion. The kernel of this action is the identity subgroup, since the only identity symmetry fixes every vertex. Formally,

$$\ker(G \times S) = \{g \in G \mid g \cdot s = s \ \forall s \in S\}.$$

Proposition 2.2.3 Kernel lies in stabilizers

For any group action G on a set S , the kernel $\ker(G \times S) \subseteq G_s$ for any $s \in S$.

Proof: $\ker(G \times S) \subseteq G_s$: fix an arbitrary $x \in \ker(G \times S)$. By definition, $\forall s \in S, x \cdot s = s$. Then for G_s we have $x \cdot s = s$, so $x \in G_s$. Thus $\ker(G \times S) \subseteq G_s \subseteq G$. Since $\ker(G \times S) \leq G$, $\ker(G \times S) \subseteq G_s$. This is true by the containment lemma. ☺

Claim 2.2.1 Containment lemma

If $H \leq G$, $K \leq G$ and $H \subseteq K$, then $H \leq K$.

Proof: 1. **Nonempty:** $H \leq G \Rightarrow e \in H \Rightarrow H \neq \emptyset$.

2. **Closure under ab^{-1} :** for arbitrary $a, b \in H$; since $H \leq G$, the subgroup property gives $ab^{-1} \in H$; and since $H \subseteq K$, $ab^{-1} \in K$. Thus $\forall a, b \in H, ab^{-1} \in K$.

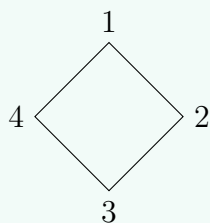
Hence $H \leq K$. ☺

The lemma feels trivial: for a subgroup $A \leq G$ to be a subgroup of K one only needs $A \subseteq K$ and closure, which A already satisfies as a subgroup of G ; the subset property is all that remains. Equivalently, any nonempty set A closed under inverses and products is a subgroup of *any* group G with $A \subseteq G$.

A reminder of the definition: a subgroup A of G is a group with $A \subseteq G$. The “subgroup test” version is a shortcut, since closure and non-emptiness (plus G being a group) already give the group axioms of A ; the remaining check is trivial and so is usually omitted.

Example 2.2.6 (2.)

The group $G = D_8$ also acts on the set A whose elements are the two unordered pairs of opposite vertices. The kernel of the action of D_8 on this set A is the subgroup $\{1, s, r^2, sr^2\}$, but for either $a \in A$, the stabilizer of a in D_8 equals the kernel of the action — for example for this square:



$$A = \{\{1, 3\}, \{2, 4\}\}.$$

Note:-

To simply check.

$\ker(G \times A) = \{g \in G \mid ga = a \forall a \in A\}$. 1 is the identity; each reflection s fixes each pair; and r^2 (a 180° rotation) and sr^2 also fix each pair. Hence $\ker = \text{Stab}(a) = \{1, s, r^2, sr^2\}$.

Definition 2.2.5: Conjugation action

Let $S = \mathcal{P}(G)$, the collection of all subsets of G . Let G act on S by conjugation: that is, for each $g \in G$ and $B \in \mathcal{P}(G)$, define

$$g \cdot B = gBg^{-1} \quad \text{where} \quad gBg^{-1} = \{gbg^{-1} \mid b \in B\}.$$

Under this action it's easy to check that $N_G(A)$ is precisely the stabilizer of A in G , i.e. $N_G(A) = G_A$, where $A \in \mathcal{P}(G)$; so $N_G(A)$ is a subgroup of G .

Let the group $N_G(A)$ act on $S = A$ by conjugation: for all $g \in N_G(A)$, $a \in A$,

$$g \cdot a = gag^{-1}.$$

This maps A to A by the definition of $N_G(A)$, so it gives an action on A . Now

$$C_G(A) = \{g \in G \mid gag^{-1} = a \forall a \in A\},$$

and the kernel of this action is precisely $\{g \in N_G(A) \mid gag^{-1} = a \forall a \in A\}$; thus $C_G(A)$ is the kernel. Hence $C_G(A) \leq N_G(A) \leq G$.

Finally, $Z(G)$ is the kernel of G acting on $S = G$ by conjugation: $g \cdot a = gag^{-1}$ for $a \in G$. We have

$$Z(G) = \{g \in G \mid gx = xg \forall x \in G\},$$

and the kernel is

$$\{g \in G \mid g \cdot x = x \forall x \in G\} = \{g \in G \mid gxg^{-1} = x \forall x \in G\},$$

which is the same as $Z(G)$. So $Z(G) \leq G$.

Example 2.2.7 (2.)

$$Z(G) = \{g \in G \mid gx = xg \forall x \in G\}, \quad C_G(Z(G)) = \{g \in G \mid gag^{-1} = a \forall a \in Z(G)\}.$$

Since for all $a \in Z(G)$, $ax = xa$ for all $x \in G$, take $g \in G$; then $gag^{-1} = a$ (as a commutes with g). So every g lies in $C_G(Z(G))$, i.e. $G \subseteq C_G(Z(G))$. By definition $C_G(Z(G)) \subseteq G$. Hence $C_G(Z(G)) = G$.

Example 2.2.8 (6.a))

Let $H \subseteq G$, and $N_G(H) = \{g \in G \mid gHg^{-1} = H\}$. We need to show that every $h \in H$ lies in $N_G(H)$. For $h \in H$, for all $x \in H$, $h \cdot x \cdot h^{-1} \in H$ (by closure); thus $h \in N_G(H)$. For the other inclusion, H needs to be a subgroup, since we need closure. Hence $H \subseteq N_G(H)$.

Note:-

Counterexample (if H is merely a subset, not a subgroup): take $G = S_3$ and $H = \{(12), (13)\}$, which is not a subgroup. Take $h = (12) \in H$. Then

$$h \cdot (13) \cdot h^{-1} = (12)(13)(12) = (23),$$

and $(23) \notin H$, so $hHh^{-1} \neq H$ and $h \notin N_G(H)$.

Example 2.2.9 (6.b))

H is abelian $\iff H \subseteq C_G(H)$.

($\Rightarrow$) If $H \subseteq C_G(H)$, then for all $h_1, h_2 \in H$, $h_1h_2 = h_2h_1$ because $h_1 \in C_G(H)$. Thus H is abelian.

($\Leftarrow$) If H is abelian, then for all $h_1, h_2 \in H$, $h_1h_2 = h_2h_1$, so $h_1 \in C_G(H)$. Since h_1 was arbitrary, $H \subseteq C_G(H)$.

Example 2.2.10 (6.)

The stabilizer G_1 of 1 under the action of S_n on $\{1, 2, \dots, n\}$ satisfies $G_1 \leq S_n$. Indeed, G_1 depends on the action on the set $\{2, \dots, n\}$ (which has $n - 1$ elements): since each $\sigma \in G_1$

is a permutation of those $n - 1$ elements, there are $(n - 1)!$ ways to permute them, so

$$|G_1| = (n - 1)!.$$

Example 2.2.11 (2.a))

Let $H \leq G$, $A \subseteq G$, and

$$N_H(A) = \{h \in H \mid hAh^{-1} = A\}, \quad N_G(A) = \{g \in G \mid gAg^{-1} = A\}.$$

($\subseteq$) For arbitrary $h \in N_H(A)$, $hAh^{-1} = A$, so $h \in N_G(A)$ by definition; moreover $h \in H$, so $h \in N_G(A) \cap H$.

($\supseteq$) For arbitrary $g \in N_G(A) \cap H$, $gAg^{-1} = A$; since $g \in H$, $g \in N_H(A)$.

Hence $N_H(A) = N_G(A) \cap H$. Since $N_G(A) \leq G$ and $H \leq G$, $N_H(A)$ is the intersection of two subgroups, hence a subgroup.

Definition 2.2.6: Heisenberg group

Let

$$H(F) = \left\{ \begin{pmatrix} 1 & a & b \\ 0 & 1 & c \\ 0 & 0 & 1 \end{pmatrix} \mid a, b, c \in F \right\}$$

be the **Heisenberg group** over F . Its center is

$$Z(H(F)) = \{h \in H(F) \mid hz = zh \ \forall z \in H(F)\}.$$

2.3 Cyclic Groups and Cyclic Subgroups

Definition 2.3.1: Cyclic group

A group H is **cyclic** if H can be generated by a single element, i.e. $\exists x \in H$ s.t. $H = \{x^n \mid n \in \mathbb{Z}\}$ for usual multiplication; for addition $H = \{nx \mid n \in \mathbb{Z}\}$. In both cases we write $H = \langle x \rangle$ and say H is generated by x , a generator of H .

Note:-

A cyclic group may have more than one generator; for example, if $H = \langle x \rangle$ then $H = \langle x^{-1} \rangle$, since $(x^{-1})^n = x^{-n}$ and

$$\{x^n \mid n \in \mathbb{Z}\} = \{(x^{-1})^n \mid n \in \mathbb{Z}\}.$$

Example 2.3.1

Let $G = D_{2n} = \langle r, s \mid r^n = s^2 = 1, srs = r^{-1} \rangle$, $n \geq 3$, and let H be the subgroup of all

rotations of the n -gon, so $H = \langle r \rangle$. Then $|H| = n$ and the generator r of H has order n :

$$r^n = 1, \quad r^{n+1} = r, \quad r^{n+2} = r^2, \dots; \quad r^{-1} = r^{n-1}, \quad r^{-2} = r^{n-2}, \dots$$

In general, write any power of r , say r^t , in the form of r^k for $0 \leq k \leq n-1$. Using the Division Algorithm, write $t = nq + k$ with $0 \leq k < n$, so that

$$r^t = r^{nq+k} = (r^n)^q \cdot r^k = 1 \cdot r^k = r^k.$$

Note:-

D_n itself is not cyclic since it is not abelian.

Example 2.3.2

Let $H = \mathbb{Z}$ with operation $+$; then $H = \langle 1 \rangle$ (here the identity is 0). For $n \in \mathbb{Z}$, $n = n \cdot 1$, so $|H| = \infty$ and $H = \langle 1 \rangle$. Since every integer can be written as a multiple of 1 [or of -1], we also have $H = \langle -1 \rangle$.

Proposition 2.3.1 $|H| = |x|$

If $H = \langle x \rangle$, then $|H| = |x|$; specifically:

1. if $|H| = n < \infty$ then $x^n = 1$ and $1, x, x^2, \dots, x^{n-1}$ are all the distinct elements of H ;
2. if $|H| = \infty$ then $x^n \neq 1$ for all $n \neq 0$, and $x^a \neq x^b$ for $a \neq b$ (so $|x| = \infty$).

Proof: Let $|x| = n < \infty$. Then $1, x, \dots, x^{n-1}$ are distinct: if $x^a = x^b$ for $0 \leq a < b \leq n-1$, then $x^{b-a} = 1$ with $0 < b-a < n$, contradicting that n is the smallest positive power of x giving the identity. So H has at least n elements. It remains to show these are all of them; if x^t is any power of x , as above write $t = nq + k$ with $0 \leq k < n$, so

$$x^t = x^{nq+k} = x^k \in \{1, x, \dots, x^{n-1}\}.$$

Suppose $|x| = \infty$, so no positive power of x is the identity. If $x^a = x^b$ for some $a < b$, then $x^{b-a} = 1$, a contradiction. So distinct powers of x give distinct elements of H , and $|H| = \infty$. ☺

Proposition 2.3.2 Common power

Let G be an arbitrary group, $x \in G$, and $m, n \in \mathbb{Z}$. If $x^n = 1$ and $x^m = 1$, then $x^d = 1$ where $d = \gcd(m, n)$. In particular, if $x^m = 1$ for some $m \in \mathbb{Z}$, then $|x|$ divides m .

Proof: By the Euclidean Algorithm, $\exists r, s \in \mathbb{Z}$ with $d = mr + ns$. Thus

$$x^d = x^{mr+ns} = (x^m)^r (x^n)^s = 1^r \cdot 1^s = 1.$$

If $x^m = 1$, let $n = |x|$. If $m = 0$ [the result is trivial], so assume $m \neq 0$. Since some nonzero power of x is the identity, $n < \infty$. Let $d = \gcd(m, n)$; by the previous result $x^d = 1$. Since $0 < d \leq n$ and n is the smallest positive power of x giving the identity, $d = n$, i.e. $n \mid m$. ☺

Theorem 2.3.1 Cyclic groups of the same order

Any two cyclic groups of the same order are isomorphic. More specifically:

1. If $n \in \mathbb{Z}^+$ and $\langle x \rangle, \langle y \rangle$ are both cyclic groups of order n , then the map

$$\varphi : \langle x \rangle \rightarrow \langle y \rangle, \quad x^k \mapsto y^k,$$

is well defined and is an isomorphism.

2. If $\langle x \rangle$ is an infinite cyclic group, then the map

$$\varphi : \mathbb{Z} \rightarrow \langle x \rangle, \quad k \mapsto x^k,$$

is well defined and is an isomorphism.

Proof: Suppose $\langle x \rangle, \langle y \rangle$ are both cyclic groups of order n , and let $\varphi : \langle x \rangle \rightarrow \langle y \rangle$ be defined as above. We must prove φ is well defined: if $x^a = x^b$, then $\varphi(x^a) = \varphi(x^b)$.

If $x^a = x^b$, then $x^{a-b} = 1$; since $|x| = n$ we get $n \mid a - b$. Write $tn = a - b$, i.e. $a = tn + b$. Thus

$$\varphi(x^a) = \varphi(x^{tn+b}) = (y^n)^t y^b = 1^t y^b = y^b = \varphi(x^b),$$

so φ is well defined.

It remains to check that φ is a homomorphism:

$$\varphi(x^a x^b) = \varphi(x^{a+b}) = y^{a+b} = y^a y^b = \varphi(x^a) \varphi(x^b).$$

Every element of $\langle y \rangle$ has the form y^k for some $k \in \mathbb{Z}$, and x^k is a preimage of y^k , so φ is surjective. Since $|\langle x \rangle| = |\langle y \rangle| = n$, a surjection from one onto the other is a bijection, so φ is an isomorphism.

If $\langle x \rangle$ is infinite, let $\varphi : \mathbb{Z} \rightarrow \langle x \rangle$ be defined as above. By the previous proposition, $x^a \neq x^b$ whenever $a \neq b$, so φ is injective; it is surjective by the definition of a cyclic group, is well defined (no ambiguity), and $x^{a+b} = x^a x^b$, so φ is a homomorphism, hence an isomorphism. $\odot$

For each $n \in \mathbb{Z}^+$, let $\mathbb{Z}_n$ denote a cyclic group of order n (written multiplicatively); then every cyclic group of order n is isomorphic to $\mathbb{Z}_n$.

Proposition 2.3.3 Order of a power

Let G be a group, $x \in G$, and $a \in \mathbb{Z} - \{0\}$.

1. If $|x| = \infty$, then $|x^a| = \infty$.
2. If $|x| = n < \infty$, then

$$|x^a| = \frac{n}{\gcd(n, a)}.$$

3. In particular, if $|x| = n < \infty$ and a is a positive integer dividing n , then $|x^a| = \frac{n}{a}$.

Proof: (1) Suppose $|x| = \infty$ but $|x^a| = m < \infty$, for contradiction. Then $1 = (x^a)^m = x^{am}$, and $x^{-am} = (x^{am})^{-1} = 1^{-1} = 1$. Since a and m are both nonzero, am or $-am$ is positive; this contradicts the definition of the order of x being ∞ .

(2) Let $y = x^a$ and $d = \gcd(n, a)$; write $n = db$, $a = dc$ for some $b, c \in \mathbb{Z}$ with $b > 0$. Since

$d = \gcd(a, n)$, the numbers b and c are relatively prime. Let $k = |y|$.
 $(k \mid b)$: first note that

$$y^b = x^{ab} = x^{dcb} = (x^{db})^c = (x^n)^c = 1^c = 1.$$

By the previous result, $k \mid b$.

$(b \mid k)$: since $x^{ak} = y^k = 1$, by the previous result $n \mid ak$, i.e. $db \mid dck$, hence $b \mid ck$. Since $(b, c) = 1$, we get $b \mid k$. Now $b = \frac{n}{\gcd(n, a)}$ and $k = |y|$, so (2) is proved.

(3) is a special case of (2). ☺

Proposition 2.3.4 Generators of a cyclic group

Let $H = \langle x \rangle$:

1. If $|x| = \infty$, then $H = \langle x^a \rangle$ iff $a = \pm 1$.
2. If $|x| = n < \infty$, then $H = \langle x^a \rangle$ iff $\gcd(a, n) = 1$. In particular, the number of generators of H is $\varphi(n)$, where φ is the Euler φ -function (the number of integers from 1 to n that are relatively prime to n).

Proof: (1) $(\Rightarrow)$ Assume $H = \langle x^a \rangle$; by definition $H = \{x^n \mid n \in \mathbb{Z}\}$. Since $x \in H$, $x = x^{an}$ for some $n \in \mathbb{Z}$, so $x^{an-1} = 1$. But x has infinite order, so $an - 1 = 0$, i.e. $an = 1$. Since $a, n \in \mathbb{Z}$, $a = 1, n = 1$ or $a = -1, n = -1$; thus $a = \pm 1$.

$(\Leftarrow)$ Assume $a = \pm 1$. If $a = 1$, then $H = \langle x \rangle = \langle x^a \rangle$. For $a = -1$: $\{x^{-n} \mid n \in \mathbb{Z}\} = \{x^n \mid n \in \mathbb{Z}\}$ since $x^{-n} = (x^{-1})^n$; thus $\langle x \rangle = \langle x^{-1} \rangle$. Hence $H = \langle x^a \rangle \Leftrightarrow a = \pm 1$.

(2) If $|x| = n < \infty$, then by the previous theorem, since x^k are distinct for $k \in \{0, \dots, n-1\}$, the element x^a generates a subgroup of H of order $|x^a|$, which equals H iff $|x^a| = |x|$. By the order-of-a-power theorem, $|x^a| = |x|$ iff $\frac{n}{\gcd(n, a)} = n \Leftrightarrow \gcd(n, a) = 1$. Since $\varphi(n)$ is the number of $a \in \{1, \dots, n\}$ with $(a, n) = 1$, $\varphi(n)$ is the number of generators. ☺

Theorem 2.3.2 Subgroups of cyclic groups

Let $H = \langle x \rangle$ be a cyclic group.

1. Every subgroup of H is cyclic. More precisely, if $K \leq H$, then either $K = \{1\}$ or $K = \langle x^d \rangle$, where d is the smallest positive integer such that $x^d \in K$.
2. If $|H| = \infty$, then for any distinct nonnegative integers a, b , $\langle x^a \rangle \neq \langle x^b \rangle$. Furthermore, for every integer m , $\langle x^m \rangle = \langle x^{|m|} \rangle$ (where $|m|$ is the absolute value of m). So the nontrivial subgroups of H correspond bijectively to the integers $1, 2, 3, \dots$.
3. If $|H| = n < \infty$, then for each positive integer $a \mid n$ there is a unique subgroup of H of order a ; this subgroup is $\langle x^d \rangle$ where $d = \frac{n}{a}$. Furthermore, for every integer m , $\langle x^m \rangle = \langle x^{\gcd(n, m)} \rangle$, so the subgroups of H correspond bijectively with the positive divisors of n .

Proof: (1) Let $K \leq H$. If $K = \{1\}$ we are done; otherwise $K \neq \{1\}$, so there is $a \neq 0$ with $x^a \in K$. If $a < 0$, then since K is a group, $x^{-a} = (x^a)^{-1} \in K$; hence K always contains some positive power of x . Let

$$\Phi = \{b \in \mathbb{Z}^+ \text{ and } x^b \in K\}.$$

Then $\Phi \subseteq \mathbb{Z}^+$ is nonempty, so by the well-ordering principle it has a minimum d . We claim $K = \langle x^d \rangle$.

Since K is a subgroup and $x^d \in K$, we have $\langle x^d \rangle \subseteq K$. Conversely, every element of K is of the form x^a for some integer a . By the division algorithm write $a = qd + r$ with $0 \leq r < d$. Then

$$x^r = x^{a-qd} = x^a(x^d)^{-q} \in K.$$

By minimality of d , it follows that $r = 0$, i.e. $a = qd$, and so $x^a = (x^d)^q \in \langle x^d \rangle$. This gives $K \subseteq \langle x^d \rangle$, so $K = \langle x^d \rangle$. Hence every subgroup K is cyclic, of the form $\{1\}$ or $\langle x^d \rangle$.

(3) Assume $|H| = n < \infty$ and $a \mid n$. Let $d = \frac{n}{a}$; by the order-of-a-power result, $\langle x^d \rangle$ is a subgroup of order a , showing the existence of a subgroup of order a . For uniqueness, suppose K is any subgroup of H of order a (so $K \neq \{1\}$); then $K = \langle x^b \rangle$ where b is the smallest positive integer such that $x^b \in K$. By the order-of-a-power result,

$$\frac{n}{d} = a = |K| = |x^b| = \frac{n}{\gcd(n, b)}.$$

So $d = \gcd(n, b)$; in particular $d \mid b$. Since b is a multiple of d , we have $x^b \in \langle x^d \rangle$, hence $K = \langle x^b \rangle \subseteq \langle x^d \rangle$. Since $|x^d| = a = |K|$, we get $K = \langle x^d \rangle$.

Finally, for every integer m , $\langle x^m \rangle = \langle x^{\gcd(n, m)} \rangle$. Indeed, if $d = \gcd(n, m)$ then $m = dq$ for some $q \in \mathbb{Z}$, so

$$x^m = x^{dq} = (x^d)^q \in \langle x^d \rangle,$$

hence $\langle x^m \rangle \subseteq \langle x^d \rangle$. On the other hand, by the order-of-a-power result,

$$|x^m| = \frac{n}{\gcd(n, m)} = \frac{n}{d} = |x^d|,$$

so the two subgroups have the same order and therefore coincide: $\langle x^m \rangle = \langle x^d \rangle$. Since $\gcd(n, m) \mid n$, every subgroup of H arises from a divisor of n , completing the proof. $\odot$

The theorem says how many distinct subgroups there are: the number of divisors d of $|x|$. The order-of-a-power result says which a generate the whole group H (those a with $\gcd(a, n) = 1$).

If we think of the cyclic subgroup $H = \langle x \rangle$ of order n as isomorphic to $\mathbb{Z}/n\mathbb{Z}$ under addition, via $\bar{k} \mapsto x^k$, then x^a corresponds to $\bar{a} \in \mathbb{Z}/n\mathbb{Z}$. And $\langle x^a \rangle = \langle x \rangle$ iff $\bar{a}$ generates all of $\mathbb{Z}/n\mathbb{Z}$, i.e. iff $\gcd(a, n) = 1$.

Example 2.3.3 (1.)

To see the insights of the prop and the theorem clearly, consider the group $\mathbb{Z}/12\mathbb{Z}$ (of order 12):

- (a) $\mathbb{Z}/12\mathbb{Z} = \langle 1 \rangle = \langle 5 \rangle = \langle 7 \rangle = \langle 11 \rangle$;
- (b) $\langle 2 \rangle = \langle 10 \rangle$;
- (c) $\langle 3 \rangle = \langle 9 \rangle$;
- (d) $\langle 4 \rangle = \langle 8 \rangle$;
- (e) $\langle 6 \rangle$;
- (f) $\langle 0 \rangle$;

for $1 \leq n \leq 12$ — these are the distinct subgroups, one for each divisor of 12.

Example 2.3.4 (2.)

We can obtain subgroups of a group G by forming $C_G(\langle x \rangle)$ and $N_G(\langle x \rangle)$ for each $x \in G$. If $g \in G$ commutes with x , i.e. $gx = xg$, then for all $a \in \langle x \rangle$, since $a = x^r$ for some r , we have $x^r g = g x^r$, i.e. g commutes with a . Thus $C_G(x) \subseteq C_G(\langle x \rangle)$. The reverse inclusion is simpler: for $g \in C_G(\langle x \rangle)$, taking $a = x$ gives $gx = xg$, so $g \in C_G(x)$. Hence $C_G(x) = C_G(\langle x \rangle)$.

Example 2.3.5 (6.)

By the previous theorem, $\langle x^m \rangle = \langle x^{\gcd(n,m)} \rangle$; in this case $\langle a \rangle = \langle \gcd(a, 48) \rangle$. Since $48 = 2 \times 24 = 2^2 \times 12 = 2^4 \times 3$, the only distinct cyclic subgroups are

$$\langle 1 \rangle, \langle 2 \rangle, \langle 3 \rangle, \langle 4 \rangle, \langle 6 \rangle, \langle 8 \rangle, \langle 12 \rangle, \langle 16 \rangle, \langle 24 \rangle, \langle 48 \rangle.$$

Then we can write out each element in $\langle a \rangle$; for example

$$\langle 8 \rangle = \{8, 16, 24, 32, 40, 0\},$$

whose order, by the previous result, is $\frac{48}{\gcd(48,8)} = \frac{48}{8} = 6$.

Example 2.3.6 (b.)

Let $L = \text{lcm}(m, n)$. Since $n \mid L$ and $|x| = n$, we have $x^L = e$; since $m \mid L$ and $|y| = m$, we have $y^L = e$. Then

$$(xy)^L = x^L y^L = e \cdot e = e,$$

so $|xy| \mid L$.

Example 2.3.7

In S_3 , since it is not abelian, let $x = (12)$, $y = (23)$. Both have order 2, and $\text{lcm}(|x|, |y|) = 2$, but $(12)(23) = (123)$ has order 3. Since $3 \nmid 2$, this is a counter-example.